

The CD4 transmembrane GGXXG and juxtamembrane (C/F)CV+C motifs mediate pMHCII-specific signaling independently of CD4-LCK interactions

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Abstract

CD4⁺ T cell activation is driven by 5-module receptor complexes. The T cell receptor (TCR) is the receptor module that binds composite surfaces of peptide antigens embedded within MHCII molecules (pMHCII). It associates with three signaling modules (CD3γε, CD3δE, and CD3ζ) to form TCR-CD3 complexes. CD4 is the coreceptor module. It reciprocally associates with TCR-CD3-pMHCII assemblies on the outside of a CD4⁺ T cells and with the Src kinase, LCK, on the inside. Previously, we reported that the CD4 transmembrane GGXXG and cytoplasmic juxtamembrane (C/F)CV+C motifs found in eutherian (placental mammal) CD4 have constituent residues that evolved under purifying selection (Lee, et al., 2022). Expressing mutants of these motifs together in T cell hybridomas increased CD4-LCK association but reduced CD3ζ, ZAP70, and PLCγ1 phosphorylation levels, as well as IL-2 production, in response to agonist pMHCII. Because these mutants preferentially localized CD4-LCK pairs to non-raft membrane fractions, one explanation for our results was that they impaired proximal signaling by sequestering LCK away from TCR-CD3. An alternative hypothesis is that the mutations directly impacted signaling because the motifs normally play an LCK-independent role in signaling. The goal of this study was to discriminate between these possibilities. Using T cell hybridomas, our results indicate that: intracellular CD4-LCK interactions are not necessary for pMHCII-specific signal initiation; the GGXXG and (C/F)CV+C motifs are key determinants of CD4-mediated pMHCII-specific signal amplification; the GGXXG and (C/F)CV+C motifs exert their functions independently of direct CD4-LCK association. These data provide a mechanistic explanation for why residues within these motifs are under purifying selection in jawed vertebrates. The results are also important to consider for biomimetic engineering of synthetic receptors.

eLife assessment

This study provides **valuable** new insights as to how two evolutionary conserved motifs in CD4 contribute to the CD4-mediated enhancement of TCR signaling independently of the CD4-LCK interaction. The data at hand are **convincing**, even if confined to a cell line model and not substantiated in vivo and with little new mechanistic insight provided regarding the domains of CD4 shown to have significant roles in the signaling process. Without the data from primary cells it is difficult to make statements about the quantitative contribution of LCK-dependent and independent functions of CD4 in TCR signaling.

Introduction

Delineating the mechanistic principles by which multi-module receptors drive complex biological processes is important both for our broad understanding of receptor biology, and for guiding biomimetic engineering of synthetic receptors for therapeutic purposes. For example, the design of conventional single-chain chimeric antigen receptors (CARs) utilized in CAR-T cell therapy was informed by an early 1990's understanding of receptor biology and bears little resemblance to the multi-module receptors that naturally drive T cell responses. These design differences necessitate that conventional CARs integrate with the T cell intracellular signaling machinery differently than their natural counterparts, leading us to posit that fully coopting T cell functions with synthetic receptors requires that they be designed to integrate with the T cell's intracellular signaling machinery in a way that mimics the native receptors (Harris and Kranz, 2016 [DOI](#); Kobayashi et al., 2020 [DOI](#); Wu et al., 2020 [DOI](#)). Achieving this goal requires a more complete understanding of the multi-module receptors that mediate antigen-specific T cell activation. In particular, understanding how CD4⁺ T cells respond to antigen is important for informing biomimetic engineering of CARs because recent data show that redirecting CD4⁺ T cells with CARs is important for long-lived therapeutic efficacy (Melenhorst et al., 2022 [DOI](#)).

CD4⁺ T cells are driven by 5-module receptors that recognize peptide antigens embedded within MHCII (pMHCII). Each naïve CD4⁺ T cell expresses a clonotypic receptor module, called the T cell receptor (TCR), that binds specifically to unique features of composite pMHCII surfaces (Kuhns and Badgandi, 2012 [DOI](#); Kuhns and Davis, 2012 [DOI](#)). The TCR lacks intracellular signaling domains and instead assembles with three signaling modules (CD3 $\gamma\epsilon$, $\delta\epsilon$, and ζ) that have immunoreceptor tyrosine-based activation motifs (ITAMs), and other motifs, to connect TCR-pMHCII interactions to the intracellular signaling apparatus. CD4 is the coreceptor module. It binds pMHCII in a reciprocal fashion with TCR-CD3 on the outside of CD4⁺ T cells and interacts with the Src kinase, LCK, via an intracellular CQC zinc clasp motif and helix (Huse et al., 1998 [DOI](#); Kim et al., 2003 [DOI](#)). According to the TCR signaling paradigm, co-engagement of pMHCII by TCR-CD3 and CD4 positions LCK and the CD3 ITAMs in the proper spatial proximity to enable LCK phosphorylation of the CD3 ITAMs to initiate signaling (Rudd, 2021 [DOI](#)). However, when we tested this model directly we found that CD4-LCK interactions are not key determinants of CD3 ζ ITAM phosphorylation (pCD3 ζ) (Lee et al., 2022 [DOI](#)). Based on our work and that of other labs, including a study showing that C-terminally truncated CD4 lacking the motifs that mediate CD4-LCK interactions can nevertheless drive CD4⁺ T cell development, proliferation, and T-helper functions, we consider the question of how intracellular pMHCII-specific CD4⁺ T cell signaling is initiated to be unresolved (Glassman et al., 2018 [DOI](#); Horkova et al., 2023 [DOI](#); Killeen and Littman, 1993 [DOI](#); Xu and Littman, 1993 [DOI](#)).

The answer to this question lies with how the 5 modules of pMHCII receptors evolved over ~435 million years to refine pMHCII-specific signaling. We have therefore taken a systems biology approach to computationally analyze data from real world experiments performed in 99 jawed vertebrate species under a variety of conditions over the course of evolution. By computationally reconstructing the evolutionary history of CD4, we identify residues and motifs in the extracellular, transmembrane, and intracellular domains that are of predicted to be functionally important (Lee et al., 2022 [↗](#)).

We then cross-validated these findings using a well-established T cell hybridoma experimental system (58 α β ⁺ cells) wherein we performed structure-function analysis of mutants of key motifs to infer their function by evaluating the phenotypes of the mutants relative to the wildtype (WT). When we mutated the intracellular CQC clasp and IKRLH helix motifs that are known to mediate CD4-LCK interactions, we observed significant reductions in CD4-LCK association; yet we did not observe the predicted decrease in CD3 ζ ITAM phosphorylation (pCD3 ζ). In contrast, when we mutated the transmembrane GGXXG motif and cytoplasmic juxtamembrane (C/F)CV+C palmitoylation motif, we observed an increased frequency of CD4-LCK pairs that were preferentially localized to detergent soluble membrane domains. We also found evidence of reduced CD3 ζ , ZAP70, and PLC γ 1 phosphorylation (pCD3 ζ , pZAP70, and pPLC γ 1) compared with WT CD4. The reason that constituent residues of these motifs evolved under purifying selection therefore appears to be due to reduced signaling in response to pMHCII when the motifs are altered. We took these data as evidence that the GGXXG and (C/F)CV+C motifs arose in eutherians (placental mammals) to tightly regulate pMHCII-specific signal initiation and postulated that the mutant CD4 molecules might sequester LCK away from TCR-CD3 to prevent CD3 ζ phosphorylation. An alternative hypothesis is that the GGXXG and (C/F)CV+C motifs impact pCD3 ζ levels independently of CD4-LCK interactions.

In the present study we tested these hypotheses by studying the impact of GGXXG and (C/F)CV+C motif mutants individually, or together, on proximal pMHCII-specific signaling in C-terminally truncated CD4 molecules (T1). Working with T1 eliminated motifs in the intracellular domain, including those that mediate CD4-LCK association. As expected from prior work, truncating CD4 relieved CD4-LCK interactions and reduced IL-2 production. However, there was no impact on the average pCD3 ζ levels or the percent of cells with phosphorylated CD3 ζ when compared with WT CD4, despite significantly reduced levels of CD4-LCK pairs. These data provide further evidence that the core tenet of the TCR signaling paradigm, wherein CD4 recruits LCK to CD3 ITAMs initiates signaling, needs revising. Importantly, T1 molecules bearing mutations in the GGXXG and (C/F)CV+C motifs individually or together reduced IL-2 production as well as pCD3 ζ levels and other TCR-proximal signaling events. The simplest interpretation of these data is that the GGXXG and (C/F)CV+C motifs are key determinants of pMHCII-specific signaling on their own, independent of CD4-LCK interactions.

Results

The goal of this study was to evaluate if the transmembrane GGXXG and juxtamembrane (C/F)CV+C palmitoylation motifs, which are highly conserved in eutherians (placental mammals) and contain constituent residues under purifying selection (**Figure 1** [↗](#)), influence pMHCII-specific proximal signaling on their own rather than by sequestering CD4-LCK pairs away from membrane rafts. Accordingly, we used 58 α β ⁺ T cell hybridomas transduced to express the 5c.c7 TCR along with either the WT or mutant CD4 constructs described in **Table 1** [↗](#). The well-characterized 5c.c7 TCR recognizes the moth cytochrome c peptide 88-103 (MCC) in the context of mouse MHCII I-E^k. We used this experimental system because 58 α β ⁺ cells were used in seminal work to link CD4-LCK association to IL-2 production (Glaichenhaus et al., 1991 [↗](#)). Specifically, mutating or removing the

intracellular CQC clasp motif was shown in that study to reduce both CD4-LCK association and signaling output as measured by IL-2 production. We confirmed those findings in our recent study (Lee et al., 2022 [↗](#)).

WT and T1 equivalently enhance pMHCII-specific proximal signaling

Previously we reported that the C-terminally truncated mouse CD4 T1 mutant (**Table 1** [↗](#) and **Figure 1** [↗](#) arrow) reduces but does not prevent IL-2 production relative to WT CD4 despite lacking most of the intracellular domain, including the CQC clasp and IKRL motifs shown to mediate CD4-LCK association (Lee et al., 2022 [↗](#)). Because T1 maintains both the transmembrane GGXXG and juxtamembrane (C/F)CV+C motifs, we reasoned that it had utility for studying the contributions these motifs make to pMHCII-specific signaling in the absence of CD4-LCK interactions.

Accordingly, we generated 5c.c7⁺ 58αβ⁺ cells expressing WT CD4, T1, or T1 combined with our previously reported Δbind mutant that disrupts binding to pMHCII in the CD4 D1 domain. This T1^{Δbind} mutant allowed us to eliminate the contributions of CD4-pMHCII interactions as well as CD4-LCK interactions (**Table 1** [↗](#) and **Figure 2 – figure supplement 1**) (Glassman et al., 2016 [↗](#); Glassman et al., 2018 [↗](#); Parrish et al., 2015 [↗](#)). The goals were to: 1) determine if T1 significantly impairs pMHCII-specific signaling events, relative to WT, as expected because it lacks the CQC and IKRL motifs that mediate CD4-LCK interactions; 2) use T1^{Δbind} to establish the baseline above which pMHCII-dependent assembly of CD4 with TCR-CD3 contributes to signaling. The T1 mutant produced measurable IL-2 (response magnitude ~61% of WT, **Table 2** [↗](#)) that was consistently reduced across a range of MCC peptide concentrations compared to WT, whereas the T1^{Δbind} mutant did not produce IL-2. We also observed reduced pMHCII-specific endocytosis for T1 and T1^{Δbind} compared to WT (**Figure 2A** [↗](#) and **Figure 2 – figure supplement 2**). These data indicate that, in our system, extracellular CD4- pMHCII engagement is essential for IL-2 production while the CD4 intracellular domain C-terminal of R422 contributes to IL-2 production, but is not absolutely necessary in the absence of other elements of the intracellular domain.

Next, we evaluated the impact of these mutations on proximal signaling events measured by flow cytometry in response to antigen presenting cells (APCs) expressing agonist MCC:I-E^k after background subtraction of signal levels in response to APCs expressing null Hb:I-E^k (**Figure 2 – figure supplement 3**). Our analysis of five independently generated pairs of WT and T1 cell lines showed variable difference in pCD3ζ levels. For three pairs of lines, the T1 response was higher than the WT and for two it was lower (**Figure 2B** [↗](#)). The averages of the T1 response normalized to the paired WT controls showed no difference between the population. There were also no differences for the average percent of T1 cells with response levels to MCC that were above background responses to the null Hb peptide compared to their paired WT controls. Similar results were observed for pZAP70 and pPLCy1 levels (**Figure 2 – figure supplement 4**). These data, summarized in **Table 2** [↗](#), provide evidence that CD4- LCK interactions via the CQC clasp and IKRL motifs are not key determinants of pMHCII-specific early signal events in this experimental system.

In comparison, the T1^{Δbind} mutant significantly reduced pCD3ζ, pZAP70, or pPLCy1 levels compared to the T1 control but, interestingly, did not impact the percent of cells responding to agonist MCC:I-E^k (**Figure 2C** [↗](#) and **Figure 2 – figure supplement 5**). Because the antibodies used to detect pCD3ζ, pZAP70, and pPLCy1 recognize single phosphorylated tyrosines, these data indicate that CD4 enhances the number of TCR-CD3 complexes per cell with phosphorylated CD3ζ molecules at Y142 (same in human and mouse), as well as the number of phosphorylated ZAP70 and PLCy1 molecules per cell, in a manner that is dependent on TCR-CD3 and CD4 co- engagement of pMHCII but independent of the intracellular domain C-terminal of R422. A summary of these

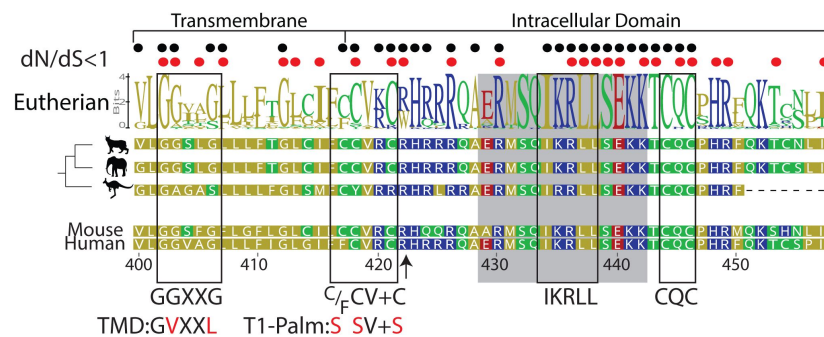


Figure 1.

Computational reconstruction of CD4 evolution.

The maximum likelihood phylogenetic tree clusters mammalian CD4 sequences. The tree highlights ancestral reconstructions of CD4 sequences from marsupials (kangaroo silhouette), Atlantogenata (elephant), and Boreoeutheria (wildcat). The logoplot of extant eutherian (Atlantogenata and Boreoeutheria) CD4 sequences show sequence conservation over evolutionary time. In these plots, the height of symbols indicates the relative frequency of each amino acid at that specific position. The mouse CD4 numbering (uniprot) is used as a reference, and residues are color-coded based on sidechain polarity. Evolutionary insertion or deletion events are indicated by dashes (-). Most recent common ancestor (MRCA) sequences are shown at each node in the tree (Node 1-4). As in our previous study (Lee et al., 2022), the ratio of synonymous (dS) and nonsynonymous (dN) substitution rates was calculated. Black dots indicate dN/dS ratios that are significantly below 1 across the entire dataset. Red dots indicate residues under purifying selection in the mammalian only dataset. Previously identified motifs are indicated by boxes, while the intracellular domain helix is shaded gray. The arrow at position 422 indicates where CD4 was truncated (CD4-T1), while TMD and T1-Palm show the mutations studied in this study.

Motif location/known function	Mutant Names	Mutated Motif	Residue Mutations
C-terminally truncated CD4	T1	Ends at R422	R422 is the last residue
Extracellular D1 C"-strand (binds pMHCII)	T1 ^{Δbind}	GKGVLR	K68D, V70D, L71S, I72D, R73S
TMD/Protein or cholesterol interactions	T1-TMD	GGxxG	G403V, G406L
Juxtamembrane/Palmitoylation	T1-Palm (2C)	CV+C	C418S, C421S
Juxtamembrane/Palmitoylation	T1-Palm (3C)	CCV+C	C471S, C418S, C421S
TMD+Palm/raft localization	T1-TP (2C)	GGxxG, CV+C	G403V, G406L, C418S, C421S
TMD+Palm/raft localization	T1-TP (3C)	GGxxG, CCV+C	G403V, G406L, C417S, C418S, C421S

Table 1.

Motifs and mutants analyzed in this study.

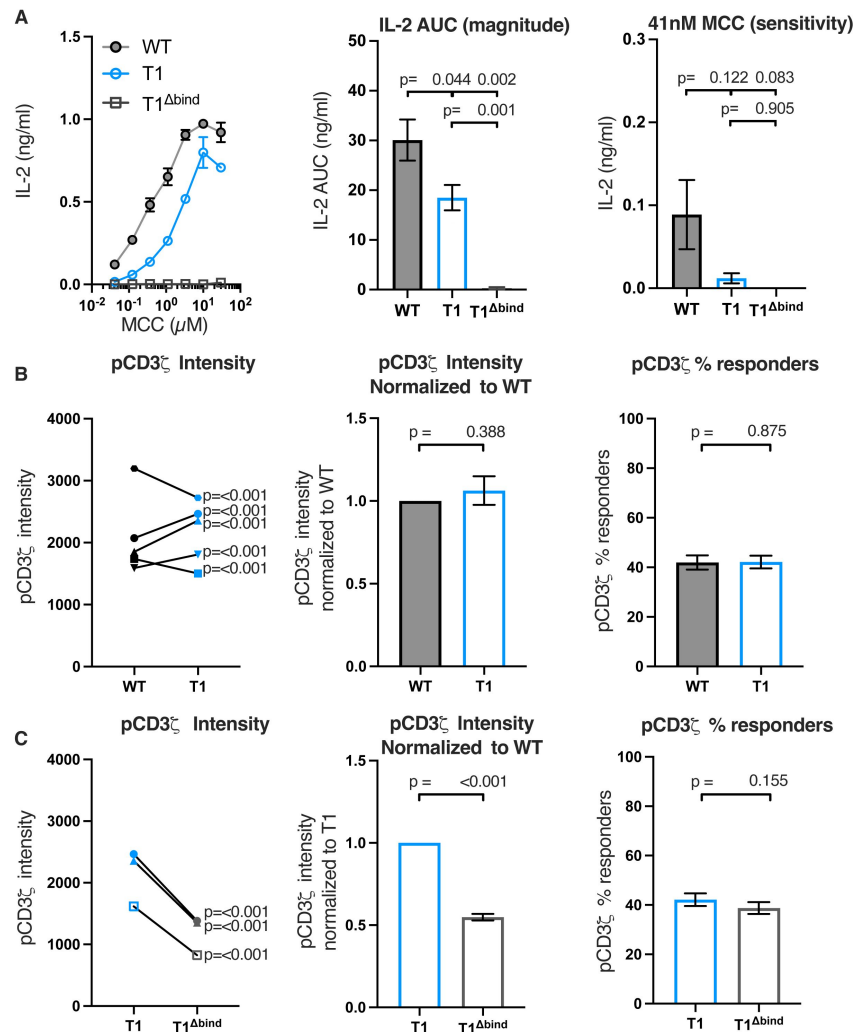


Figure 2.

Truncating CD4 does not reduce CD3 ζ phosphorylation.

(A) Representative IL-2 production is shown in response to a titration of MCC peptide from one experiment (left). Experiments were performed in triplicate and each symbol equals the mean $\pm$ SEM at that peptide concentration. AUC analysis for the dose response is shown as a measure of the response magnitude for the average of three independent experiments performed with one independently generated set of cell lines (center). The average response to a low dose (41nM) of peptide is shown as a measure of sensitivity for three independent experiments performed with one independently generated set of cell lines (right). The data are representative of those obtained with 4 (WT vs T1) or 2 (WT vs $T1^{\Delta bind}$) independently generated sets of cell lines.

(B) Phosphorylation intensity of CD3 ζ (pCD3 ζ) for WT and CD4-T1 (T1) (left), normalized pCD3 ζ intensity for WT and T1 (center), and average % responders of pCD3 ζ for WT and T1 (right). Five independently generated cell lines (biological replicates) were tested for WT and T1. For phosphorylation intensity, each pair of lines (connected symbols) was tested in three independent experiments. Data were analyzed and collected as in Figure 2 – figure supplement 3. One-way ANOVA was performed with a Dunnett's posttest using GraphPad Prism9. For normalized intensity, all individual mutant cell line values were normalized to their paired control values. Average percent responders are presented. Bars represent the mean $\pm$ SEM. One-way ANOVA with paired comparisons was performed with a Sidak's posttest for specific comparisons of normalized values using GraphPad Prism9. All generated cells lines were considered for the multiple comparisons.

(C) Phosphorylation intensity of pCD3 ζ for T1 and $T1^{\Delta bind}$ (left), normalized pCD3 ζ intensity for T1 and $T1^{\Delta bind}$ (center), and % responders of pCD3 ζ for T1 and $T1^{\Delta bind}$ (right) were performed and analyzed as in B, with the exception that the open symbols represent data from a single experiment whereas the closed symbols represent aggregate data from three independent experiments. Two-tailed t tests were performed to compare the single T1 vs $T1^{\Delta bind}$ samples as no other samples were collected in parallel.

Mutant Names	IL-2 magnitude	IL-2 sensitivity	Normalized pCD3ζ intensity	Normalized % Responders pCD3ζ	Normalized pZap70 intensity	Normalized % Responders pZap70	Normalized pPlcy1 intensity	Normalized % Responders pPlcy1	Lck DRM AUC	Lck Total AUC	Lck DSM AUC	CD4 expression	CD4 endocytosis
WT	100	100	100	100	100	100	100	100	100	100	100	100	100
T1	61.46	13.52	106.30	95.20	106.10	102.20	104.40	112.60	28.81	19.22	16.08	110.09	14.30

Table 2.

All values presented as percent of WT control (truncation average/WT average x 100) for normalized values.

results can be found in [Table 3](#). A key takeaway is that the average pCD3 ζ levels for three independently generated T1^{Abind} mutants was 54.86% of the T1 control (p<0.001). CD3 ζ phosphorylation therefore occurs upon TCR engagement of cognate pMHCII in the absence of CD4-pMHCII interactions, and CD4 engagement of pMHCII increases the number of phosphorylated TCR-CD3 complexes. Importantly, the pCD3 ζ levels observed in the T1^{Abind} mutants is subthreshold for IL-2 production at the bulk population level. These results allow us to conclude that pMHCII-specific baseline pCD3 ζ levels achieved in our system in the absence of CD4-pMHCII interactions are approximate half of that achieved when CD4 can interact with pMHCII on the outside of the cell. Furthermore, consistent with the kinetic discrimination model of T cell activation, CD4-pMHCII interactions on the outside of the cell appear to be essential to push pCD3 ζ levels above the threshold level needed for IL-2 production ([Rabinowitz et al., 1996](#)).

The GGXXG and (C/F)CV+C motifs influence IL-2 production independently of LCK

Having established that T1 contributes to early pMHCII-specific signaling (e.g. pCD3 ζ) as well as signaling output (i.e. IL-2), we generated 5c.c7⁺ 58 α β ⁺ cells expressing WT, T1, or T1 bearing mutants of the GGXXG and/or (C/F)CV+C motifs ([Table 1](#) and [Figure 3 – figure supplement 1](#)). To study the function of the GGXXG motif alone, we used our previously described TMD mutant (GGXXG to GVXXL) wherein bulky side chains replace the glycines that compose a flat surface on the transmembrane helix that could mediate protein:protein or protein:cholesterol interactions ([Fessler, 2016](#); [Parrish et al., 2015](#); [Song et al., 2014](#); [Teese and Langosch, 2015](#); [Wacker et al., 2013](#)). We called these mutants T1-TMD. To test the function of the palmitoylation motif, which contains the core CVRC (418-421) sequence in mouse and humans, we made note of a broader (C/F)CVRC motif in eutherians wherein the majority of ortholog sequences had CCVRC, as found in mouse, and a minority had FCVRC as found in humans. Importantly, position 417 is under purifying selection for all extant CD4 orthologs we analyzed by the fixed effects likelihood (FEL) method ([Figure 1](#)), indicating that this residue evolved under purifying selection due to functional importance ([Lee et al., 2022](#)). We therefore made T1-Palm (3C) mutants wherein we mutated all three cysteines to serines as well as T1-Palm (2C) mutants where only the core cysteines common to mouse and humans were mutated to serines. Finally, we made mutants containing both the TMD and Palm mutants that we called T1-TP(3C) or T1-TP(2C).

To confirm that truncating CD4 relieved CD4-LCK association, and evaluate if the mutations impacted any residual CD4-LCK interactions that may occur due to colocalization in particular membrane domains, we performed sucrose gradient analysis of detergent lysates of WT, T1, T1-TMD, T1-Palm(2C), T1-Palm(3C), T1-TP(2C) and T1-TP(3C). We observed significantly reduced CD4-LCK association in detergent resistant membranes (DRMs, aka membrane rafts) and detergent soluble membranes (DSMs) for T1 compared to WT ([Figure 3 – figure supplement 2](#)) ([Pike, 2006](#)). The mutants did not impact residual CD4-LCK association in the DRMs or DSMs.

Furthermore, the total LCK signal associated with CD4 was significantly lower for T1 compared with the WT, and the residual association was not impacted by the mutants ([Table 3](#) and [Figure 3 – figure supplement 3](#)). We therefore conclude that, in our experimental system, any functional differences between the T1-TMD, T1-Palm(2C/3C), or T1-TP(2C/3C) mutants and the T1 control is a function of the motif we are studying in the absence of direct CD4-LCK interactions.

Next, we measured IL-2 production by WT, T1, and CD4 mutant cells to evaluate the impact of the mutations on signaling output in response to MCC. Here again, T1 was slightly lower than WT, while the T1-TMD reduced the IL-2 response magnitude and sensitivity relative to the T1 control ([Figure 3A](#)). This was expected given our prior work showing that the TMD mutant reduced IL-2 production driven by a shorter CD4 truncation mutant (CD4T, mutated at R420) ([Parrish et al., 2015](#)). Mutating the GGXXG or (C/F)CV+C motifs individually or together did not impact pMHCII-

Mutant Names	IL-2 magnitude	IL-2 sensitivity	Normalized pCD3 ζ intensity	Normalized % Responders pCD3 ζ	Normalized pZap70 intensity	Normalized % Responders pZap70	Normalized pPcyl1 intensity	Normalized % Responders pPcyl1	Lck DRM AUC	Lck Total AUC	Lck DSM AUC	CD4 expression	CD4 endocytosis
T1	100	100	100	100	100	100	100	100	100	100	100	100	100
T1 Δ bind	1.53	0.00	54.86	91.39	71.96	111.9	74.32	106.8	ND	ND	ND	73.93	38.71
TMD	53.54	17.05	76.68	94.23	88.36	98.77	88.24	107.40	62.36	92.97	110.94	131.10	3.63
Palm (2C)	57.99	116.07	76.35	94.09	86.17	93.03	92.09	98.09	274.03	139.40	130.91	80.30	-98.00
TP (2C)	33.11	0.19	62.06	85.61	79.79	75.58	82.57	83.34	121.39	113.03	114.98	96.54	47.50
Palm (3C)	27.84	0.19	71.09	94.76	86.32	76.82	80.37	88.62	137.57	94.34	92.07	87.58	6.99
TP (3C)	31.19	45.53	62.49	84.60	80.16	66.92	70.91	82.81	86.84	101.20	102.74	79.59	-54.69

Table 3.

All values presented as percent of T1 control (mutant average/T1 average x 100) for normalized values.

induced TCR endocytosis relative to the T1 control, and activation-induced CD4 endocytosis was inhibited for all truncated CD4 molecules (**Figure 3 – figure supplement 4A and 4B**). The T1-Palm(2C), T1-TP(2C), T1-Palm(3C), and T1-TP(3C) all reduced IL-2 production relative to T1, indicating that they also influence the LCK-independent contribution of CD4 to pMHCII-specific signaling (**Figure 3B**). Of note, the T1-TP(2C) IL-2 response magnitude was lower than that of T1-Palm(2C) alone, consistent with what we previously observed with the full-length versions of these CD4 mutants (**Figure 3B**) (Lee et al., 2022). Furthermore, the T1-Palm(3C) response magnitude was lower than that of T1-Palm(2C), suggesting that the additional cysteine contributes to signaling. Together, these data indicate that the GGXXG and (C/F)CV+C motifs contribute to the signaling cascade that leads to pMHCII-specific IL-2 production when CD4 does not directly interact with LCK via the intracellular CQC clasp and IKRL motifs. These data do not tell us where in the signaling pathway the contributions are made.

The CD4 GGXXG and (C/F)CV+C motifs are key determinants of early signaling

To evaluate if the GGXXG and (C/F)CV+C motifs influence early signaling events, we measured pCD3 ζ , pZAP70, and pPLC γ 1 levels for four independently generated T1-TMD lines, three independently generated T1-Palm(2C) lines, and three independently generated T1-TP(2C) lines relative to their matched T1 controls in response to APCs expressing MCC:I-E^k (**Figure 4 and Figure 4 – figure supplement 1 and 2, and Table 3**). For the T1-TMD, T1-Palm(2C), and T1-TP(2C) mutant lines we observed lower average levels of pCD3 ζ , pZAP70, and pPLC γ 1 relative to the T1 controls. These data indicate that the GGXXG and (C/F)CV+C motifs contribute to CD4's ability to enhance proximal signaling in the absence of CD4-LCK interactions.

Of note, when we considered the average fold change of the means of the independently generated mutant cell lines relative to their paired T1 controls, we included the mean of the fold change of the T1^{Abind} mutant cells from **Figure 2C** relative to their paired T1 controls for comparison (dotted line, **Figure 4B** and **Figure 4 – figure supplement 1B and 2B**). Doing so provides visual boundaries within which to consider the contribution that CD4 T1, which can bind pMHCII, makes to proximal signaling in relationship to T1^{Abind} (dotted line) that cannot bind pMHCII. These boundaries then provide an additional guide for evaluating the impact of the mutations on proximal signaling. For example, the T1-TMD and T1-Palm(2C) showed reduced pCD3 ζ levels midway between T1 and T1^{Abind}, which equated to a reduced contribution to pCD3 ζ levels made by these mutants to ~48% of the contribution to pCD3 ζ levels made by T1, while the T1-TP(2C) contributed ~16% of the pCD3 ζ levels contributed by T1. The double mutant therefore nearly eliminated the contribution made by T1 to pCD3 ζ levels. Although these are relative comparisons, the data further suggest a role for the GGXXG and (C/F)CV+C motifs in contributing to CD4-mediated pMHCII-specific signaling in the absence of CD4-LCK interactions.

Because we had found that T1^{Abind} cells failed to produce IL-2 in response to high peptide concentrations in our dose response experiments (**Figure 1A**), whereas the T1-TP(2C) mutant cells made IL-2 in response to high peptide concentrations (**Figure 3B**), we found it interesting that the early signaling responses we measured to APCs expressing high ligand densities of tethered pMHCII were similar between the T1^{Abind} and T1-TP(2C) cells. We therefore evaluated IL-2 production of the T1, T1^{Abind}, and T1-TP(2C) in response to the APCs used in these signaling studies to ask if they followed the same pattern of responses to high peptide doses in our prior experiments. We found that the T1-TP(2C) cells made approximately half as much IL-2 as the T1 controls, whereas IL-2 production by the T1^{Abind} cells was negligible (**Figure 4 – figure supplement 3**). These data indicate that the ability of CD4 to interact with pMHCII can partially overcome the loss of function of the GGXXG and (C/F)CV+C motifs at high ligand densities with respect to a downstream signaling output, such as IL-2 production, even if the early signaling events measured at 2 minutes after pMHCII-engagement were similar.

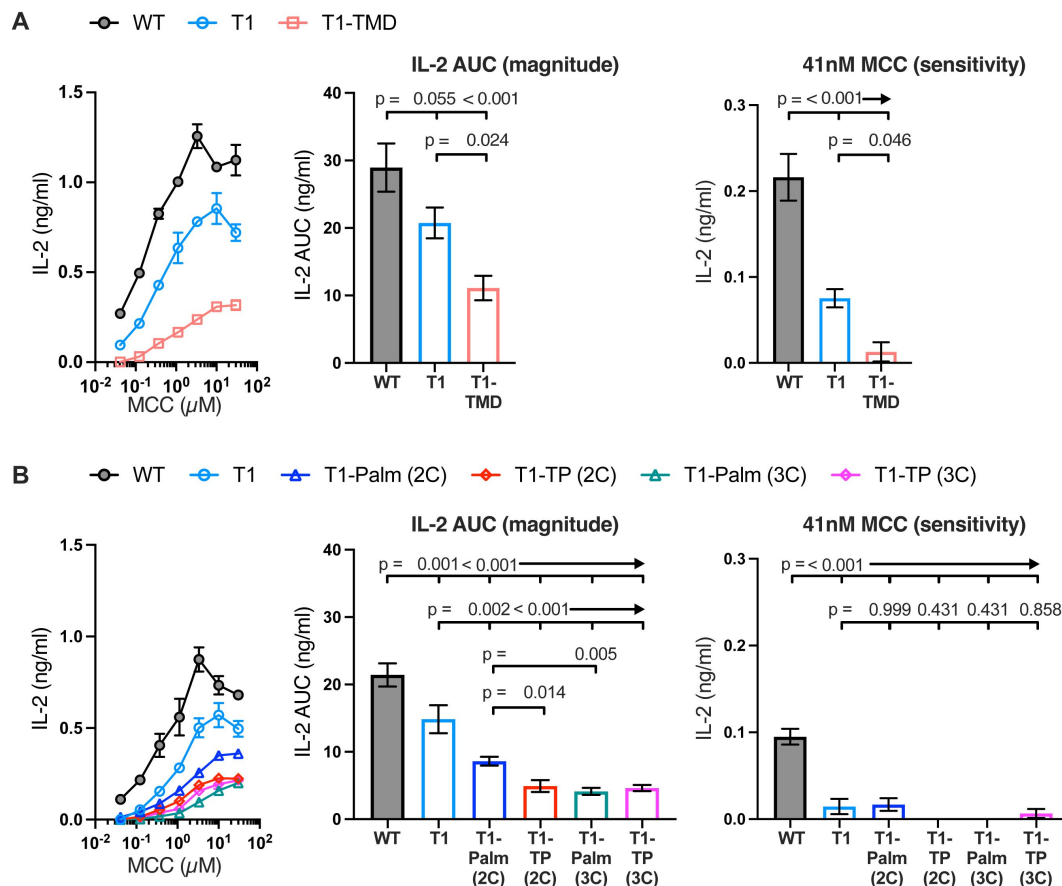


Figure 3.

The GGXXG and CV+C motifs are key determinants of CD4 function.

(A, B) Representative IL-2 production is shown in response to a titration of MCC peptide from one experiment (left). Experiments were performed in triplicate and each symbol equals the mean $\pm$ SEM at that peptide concentration. AUC analysis for the dose response is shown as a measure of the response magnitude for the average of three independent experiments performed with one independently generated set of cell lines (center). The average response to a low dose (41nM) of peptide is shown as a measure of sensitivity for three independent experiments performed with one independently generated set of cell lines (right). Results are representative of those obtained with at least three independently generated sets of cell lines. One-way ANOVA was performed with a Dunnett's posttest for comparisons with WT and T1 samples, and a Sidak's posttest for comparisons between selected samples.

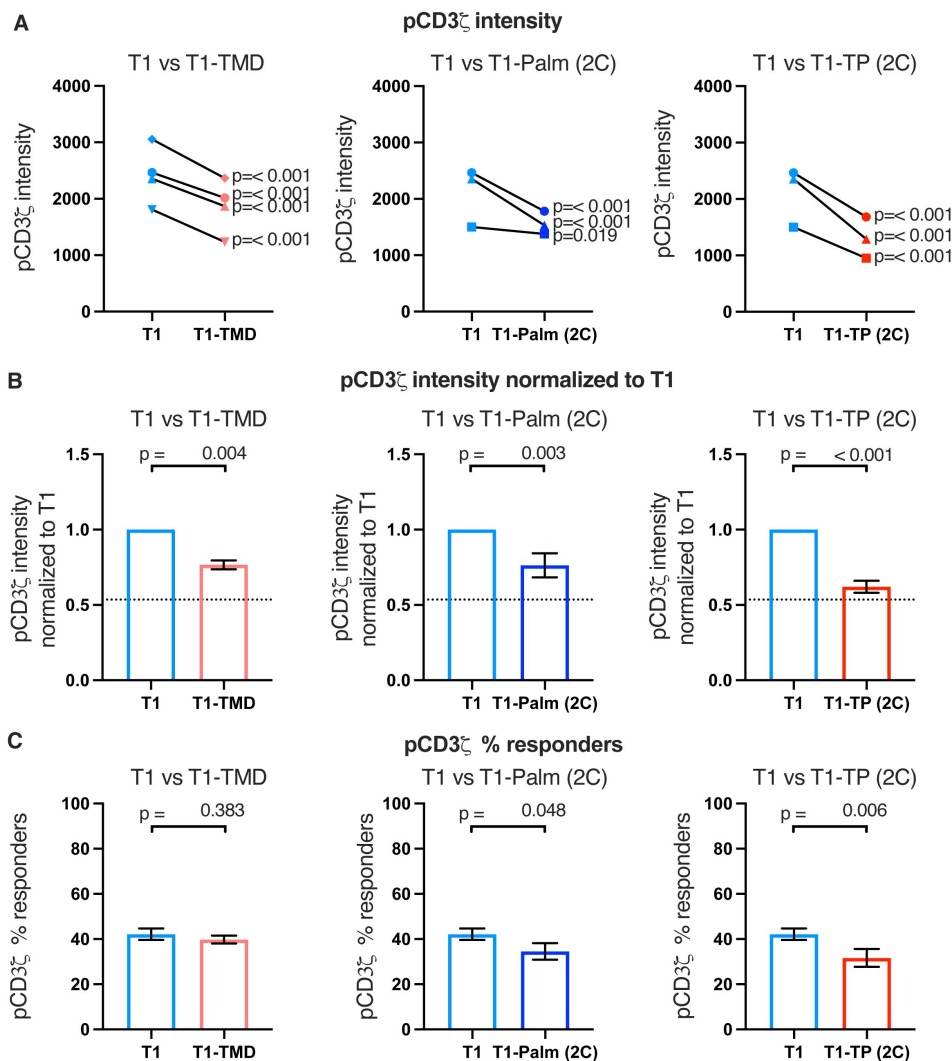


Figure 4.

The GGXXG and CV+C motifs reduce pCD3 ζ levels.

(A) Phosphorylation intensity of CD3 ζ for T1 and T1-TMD (left), T1 and T1-Palm (2C) (center), and T1 and T1-TP (2C) (right) are shown for independently generated pairs of (connecting line) T1 and T1-TMD (left), T1 and T1-Palm (2C) (center), and T1 and T1-TP (2C) (right) cell lines. Each pair of lines (connected closed symbols) was tested in three independent experiments (technical replicates). The data from those experiments was aggregated, and the symbols represent the mean intensity of the aggregated pCD3 ζ intensity values. One-way ANOVA was performed with a Dunnett's posttest using GraphPad Prims 9.

(B) Data for each cell line in **A** are shown as the average pCD3 ζ intensity for all T1-TMD (left), T1-Palm (2C) (center), and T1-TP (2C) (right) cell lines normalized to their paired T1 control. Dotted line is the normalized pCD3 ζ intensity for T1^{Abind} as a visual reference point for the contributions of CD4-pMHCII interactions. Bars represent the mean $\pm$ SEM. One-way ANOVA was performed with a Sidak's posttest for specific comparisons using GraphPad Prims 9.

(C) The average % responders for phosphorylation of CD3 ζ is shown for T1-TMD (left), T1-Palm (2C) (center), and T1-TP (2C) (right) compared to the average of their paired T1 control lines.

Bars represent the mean $\pm$ SEM. One-way ANOVA was performed with a Sidak's posttest for specific comparisons using GraphPad Prims 9.

Finally, we considered the percent of responders (i.e., those cells that signaled above background) for the mutant cell lines. We observed small decreases in responding T1-TP(2C) cells for pCD3 ζ and pZAP70 relative to the control, while pPLCy1 trended lower for this mutant. These data indicate that mutating the GGXXG and (C/F)CV+C motifs together in the absence of the intracellular domain can reduce the number of cells that initiate pMHCII-specific signaling (**Figure 4C** and **Figure 4 – figure supplement 1C and 2C**). We did not observe this phenotype in our prior study when the intracellular domain was present, suggesting this is unique to the absence of the intracellular domain (Lee et al., 2022).

We also measured early signaling for three independently generated T1- Palm(3C) and T1-TP(3C) mutant cell lines compared to their paired T1 controls. For all lines, pCD3 ζ , pZAP70, and pPLCy1 levels were significantly lower than the controls, which was reflected in the normalized fold change (**Figure 5A** and **Figure 5B** and **Figure 5 – figures supplement 1A, 1B, 2A, 2B**). The fold change for the T1-TP(3C) levels relative to the T1 was similar to T1-TP(2C) (**Figure 5B** and **Figure 5 – figures supplement 1B and 2B and Table 3**). These data provide further evidence that the GGXXG and (C/F)CV+C motifs together are key determinants of the LCK-independent contributions that CD4 makes to amplifying pMHCII-specific proximal signaling.

Finally, the percent responders were lower for pCD3 ζ and pZAP70 for T1-TP(3C) compared to WT (**Figure 5C** and **Figure 5 – figures supplement 1C and 2C**). These data are consistent with the idea that, in the absence of the intracellular domain, mutating these motifs influences initiation of key signaling events. See **Table 3** for a summary of the data.

Discussion

Our computational reconstruction of CD4 evolution provided access to a subset of results from experiments that Nature performed over ~435 million years in a greater variety of jawed vertebrates, in the wild, than could be achieved with mouse or human samples in the lab (Lee et al., 2022). It allowed us to identify residues that were selected under purifying selection, and are thus predicted to be functionally significant, because acquiring a mutation at these residues would affect fitness and results in a failure to propagate the change to future progeny. Our structure-function analysis of these residues in 58 $\alpha\beta^-$ cells, which we continued in this study, provided a second system to cross-validate the functional significance of particular residues and allowed us to start providing mechanistic insights into why these residues are important.

The data presented here are not consistent with predictions of the prevailing model in which CD4 recruits LCK to CD3 ITAMs to initiate signaling. Importantly, this study builds upon our prior work by showing that while reducing CD4-LCK interactions reduces pMHCII-specific IL-2 production in our system, either by mutating the CQC clasp or removing the CD4 intracellular domain entirely, this cannot be attributed to reductions in pCD3 ζ levels as predicted by the TCR signaling paradigm (Lee et al., 2022). It is notable that, in the absence of CD4-pMHCII interactions (T1^{Abind} mutant), pCD3 ζ levels were roughly half of those measured when CD4-pMHCII interactions occurred (T1 mutant). We take these data together as evidence that there is a threshold for the number of phosphorylated CD3 ζ molecules, and/or the duration of signaling, that is required to drive IL-2 production in our system. We also conclude that CD4 binding to pMHCII functions to increase the number of TCR-CD3 complexes per cell that experience CD3 ζ phosphorylation to reach that threshold and/or sustain signaling. Furthermore, CD4-pMHCII interactions similarly impacts pZAP70 and pPLCy1 levels, albeit to a lesser extent for the latter.

A logical extension of these conclusions is that there is a density of agonist pMHCII below which CD4 binding to pMHCII is essential for signal initiation. This idea is supported by findings that CD4 is necessary for signal propagation and amplification to the calcium mobilization step in response to fewer than 25 agonist pMHCII (Irvine et al., 2002). Attempting to accurately measure the

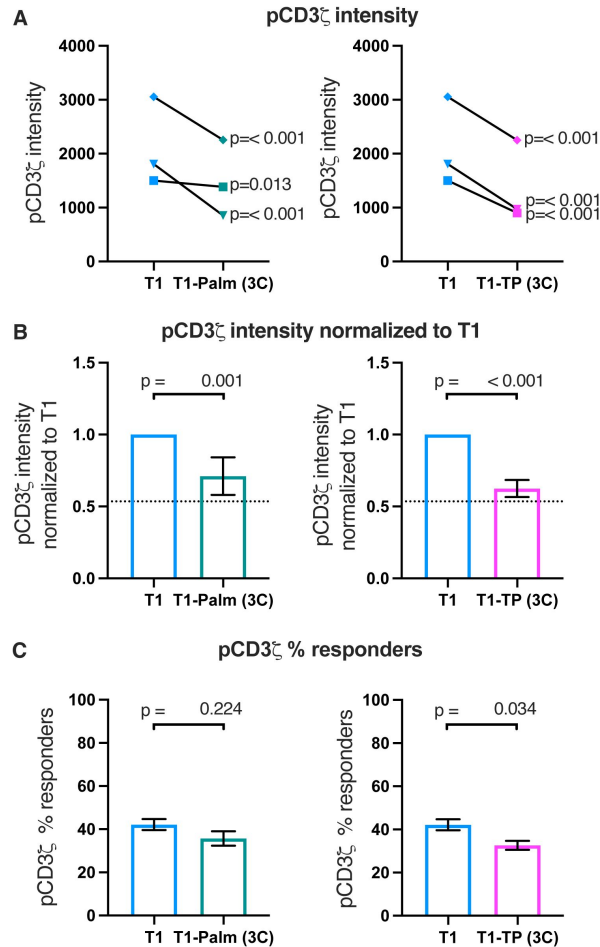


Figure 5.

The (C/F)CVRC motifs reduces proximal TCR-CD3 signaling.

(A) Phosphorylation intensity of CD3 ζ for T1 and T1-Palm (3C) (left) and T1 and T1-TP (3C) (right) are shown for independently generated pairs of (connecting lines) T1 and T1-Palm (3C) (left) and T1 and T1-TP (3C) (right) cell lines. Analysis was performed as in [Figure 4A](#).

(B) Data for each mutant cell line in **A** are shown as the average pCD3 ζ intensity values for T1-Palm (3C) (left) and T1-TP (3C) (right) normalized to their paired T1 controls. The dotted line represents the normalized phosphorylation CD3 ζ intensity for T1 Δ bind as a visual reference for the contributions of CD4-pMHCII interactions. Analysis was performed as in [Figure 4C](#).

(C) Average % responders for phosphorylation of CD3 ζ is shown for T1-Palm (3C) (left) and T1-TP (3C) (right) are shown compared to the average of their paired T1 control. Analysis was performed as in [Figure 4C](#).

contributions of CD4 to ITAM phosphorylation, with or without mutant motifs, at very low densities of pMHCII would be challenging and is technically beyond the scope of this study. Nevertheless, in this and our prior publication we did not observe the expected decreases in pCD3 ζ levels when CD4-LCK interactions were reduced by mutating the CQC clasp motif, the IKRL motif, or by eliminating the intracellular domain. Furthermore, we found in our prior study that IL-2 production increased significantly when we mutated the full intracellular helix, the helix and CQC clasp together, or the IKRL motif alone. These direct tests of the TCR signaling paradigm are inconsistent with predictions of the model. As such, they suggest that the contributions of CD4 to signal initiation and amplification at the early timepoint we measured are independent of direct CD4-LCK interactions.

Because the combined impact of mutating the GGXXG and (C/F)CV+C motifs reduced pCD3 ζ levels to ~16% of the contribution of CD4-pMHCII engagement, even when direct CD4-LCK interactions were absent, we can conclude that the GGXXG and (C/F)CV+C motifs are key determinants that help CD4 drive signaling above the threshold required for IL-2 production. Our data also provide evidence that mutating the GGXXG and (C/F)CV+C motifs together can reduce the frequency of cells experiencing CD3 ζ and ZAP70 phosphorylation in the absence of the intracellular domain. Given that palmitoylation is rapidly reversible, these data suggest that the switch-like function of the palmitoylation motif may influence signal initiation. Interestingly, unlike CD4-pMHCII interactions, these motifs are not essential for signaling output at high ligand densities as we observed low levels of IL-2 produced in cells bearing mutants of these motifs with increasing doses of agonist pMHCII. Binding of CD4 and TCR-CD3 to pMHCII can therefore amplify pMHCII-specific signaling without the contributions of the GGXXG and (C/F)CV+C motifs, or motifs in the intracellular domain, albeit at lower levels than if these motifs are intact. This could be due to the contributions of additional unidentified motifs in the transmembrane domain, or to the increase in TCR-CD3 dwell time on pMHCII that is mediated by the CD4 extracellular domain (Glassman et al., 2018 [DOI](#)).

Given that these motifs were essential for IL-2 production at low ligand densities, a logical extension of these data is that there is an agonist pMHCII density below which these motifs are essential for the initiation of early signaling events.

The data from our current study indicate that the GGXXG and (C/F)CV+C motifs do not influence early signaling events by regulating the ability of CD4 to either recruit LCK to, or sequester it away from, the TCR-CD3 ITAMs as we previously postulated (Lee et al., 2022 [DOI](#)). Instead, they work independently of CD4-LCK interactions to enhance pMHCII-specific signaling. It is tempting to speculate that because these motifs can regulate CD4 membrane domain localization and, because in other proteins palmitate moieties can sandwich cholesterol against the flat surface of a GG patch on a transmembrane domain helix, these motifs may allow CD4 to regulate local concentration of cholesterol or cholesterol sulfate around TCR-CD3, perhaps by taking such molecules away from TCR-CD3 to stabilize allosteric changes associated with signaling (Chen et al., 2022 [DOI](#); Fessler, 2016 [DOI](#); Gil et al., 2002 [DOI](#); Lee et al., 2015 [DOI](#); Song et al., 2014 [DOI](#); Swamy et al., 2016 [DOI](#); Teese and Langosch, 2015 [DOI](#); Wacker et al., 2013 [DOI](#); Wang et al., 2016 [DOI](#)). These CD4 motifs may also impact the accessibility of ITAMs for LCK phosphorylation by changing the local membrane environment around TCR-CD3 upon reciprocal engagement of pMHCII (Aivazian and Stern, 2000 [DOI](#); Xu et al., 2008 [DOI](#)). Finally, it is worth considering that, in our prior study, we found G402 of the GGXXG motif has covaried over evolutionary time with L438 of the intracellular helix IKRL motif that has inhibitory function, suggesting that the function of these motifs are co-evolving (Lee et al., 2022 [DOI](#)). In support of this idea, we reported that the signal enhancing activity of the GGXXG and (C/F)CV+C motifs together counterbalanced the inhibitory activity of the IKRL motif with regards to IL-2 production. It is unclear at this point if this counterbalancing is simply additive of individual motif functions, or if there is a functional interplay of the motifs when the intracellular domain of CD4 is present. Delineating the mechanistic details by which the GGXXG and (C/F)CV+C motifs increase signaling on their own, or when combined, and how their function integrates with those of other motifs will be the subject of future investigations.

Altogether, the data in this and our preceding study provide direct evidence that complement indirect results indicating that pMHCII-specific signaling is not initiated, or dependent on, CD4 recruitment of LCK kinase activity to the CD3 ITAMs (Glassman et al., 2018 [DOI](#); Horkova et al., 2023 [DOI](#); Killeen and Littman, 1993 [DOI](#); Lee et al., 2022 [DOI](#); Xu and Littman, 1993 [DOI](#)). Recent work on CD8 suggest that this coreceptor also does not recruit LCK to phosphorylate CD3 ITAMs (Casas et al., 2014 [DOI](#); Wei et al., 2020 [DOI](#)). In total, these findings suggest that the TCR signaling paradigm needs revising as it pertains to coreceptor-LCK interactions. For CD4, the existing evidence suggest that reciprocal binding of CD4 and TCR-CD3 enables free LCK and FYN to initiate signaling by phosphorylating the ITAMs and ZAP70 (Glassman et al., 2016 [DOI](#); Glassman et al., 2018 [DOI](#); Horkova et al., 2023 [DOI](#); Lee et al., 2022 [DOI](#); Salmond et al., 2009 [DOI](#); van Oers et al., 1996 [DOI](#)). The contribution of the current study is in showing the GGXXG and (C/F)CV+C motifs work together to enhance these early signaling events independently of direct CD4-LCK interactions.

Finally, it is interesting to consider what the primary purpose of CD4-LCK interactions via the CQC clasp and IKRL motifs are if not to recruit LCK kinase activity to the CD3 ITAMs. Data from two studies suggest that LCK makes a kinase-independent contribution to pMHCII-specific responses; it is therefore plausible that the phenotype reported for CD4 when the CQC clasp is mutated is due to relieving the kinase-independent scaffolding function of LCK (Horkova et al., 2023 [DOI](#); Xu and Littman, 1993 [DOI](#)). In addition, CD4-LCK interaction via the CQC clasp motif and the intracellular helix will prevent the helix from interacting with other partners, thus mutating the CQC clasp would favor helix interactions with other putative partners (Lee et al., 2022 [DOI](#)).

Finally, the CQC clasp is thought to play a role in positioning CD4 in proximity to LAT, either through direct or indirect interactions, which could contribute to the phenotype of CQC clasp mutants (Bosselut et al., 1999 [DOI](#); Lo et al., 2018 [DOI](#); Lo et al., 2019 [DOI](#)). If CD4-LCK and CD4-LAT interactions are mutually exclusive, and both require the CQC clasp, then mutating this motif would have a greater impact than just relieving CD4-LCK interactions (Bosselut et al., 1999 [DOI](#)). These questions remain outstanding, as do questions concerning how the intracellular helix regulates pMHCII-specific signaling.

In closing, the broader implications of the data presented in our current and prior evolution-structure-function studies provide fertile ground for future directions beyond those mentioned above. Our approach allows us to identify functionally important residues/motif and then interrogate basic mechanistic principles of their function that should hold true across CD4⁺ T cell subsets since the biochemical and biophysical bases for interactions between CD4 and its interacting partners (i.e. other proteins or membrane components) should be the same in 58αβ⁺ cells as in thymocytes or in different CD4⁺ T cells. However, different CD4⁺ T cell populations will have different levels of expression of relevant proteins (e.g. interacting partners or modifying enzymes), and may even have differences in lipid composition, such that the impact of the motifs we have identified and studied to date may lead to different outcomes in different CD4⁺ T cell subsets (<http://immpres.co.uk/> [DOI](#)) (Tuosto and Xu, 2018 [DOI](#)). For example, differences in expression of enzymes with switch-like functions, such as those that add or remove palmitate to the (C/F)CV+C motif, or those that phosphorylate or dephosphorylate the serines in the intracellular helix that regulate CD4-LCK interactions and inhibitory activity, may vary between naïve CD4⁺ T cells, different Th subsets, or Tregs to differentially tune the activity of these motifs and their impact on pMHCII-specific signaling. Now that we have identified residues and broader motifs that have proven to be functionally important *in vivo* over 435 million years of evolution, while also providing mechanistic insights as to why the residues are important, there is utility in evaluating the impact of mutant CD4 mice on thymocyte development, naïve CD4⁺ T cell activation and differentiation, and the execution of Th and Treg effector functions. In so doing, we will gain additional insights into why there is a fitness cost if a mutation is acquired in these residues. Such knowledge will increase our fundamental understanding of CD4⁺ T cell biology and will also be critical for biomimetic engineering of synthetic receptors that can redirect CD4⁺ T cell activity for therapeutic purposes.

Materials and methods

Evolutionary analyses

Available CD4 orthologs were identified through reciprocal blast-based searches and downloaded from GenBank. BLAST may not only identify orthologs, so additional criteria were used to include putative orthologous CD4 sequences in our analyses: presence of a domain structure consisting of four extracellular Ig domains followed by a transmembrane domain and a C-terminal intracellular domain, including the presence of the LCK binding clasp (CxC). Sequences that were shorter, contained frameshift mutations, or displayed high sequence variability were excluded from the analysis. For the current study, teleost fish were considered to be the oldest living species that contain a CD4 molecule given that a CD4 ortholog was not reported in the elephant shark (*Callorhynchus milii*), although future analyses of other cartilaginous fishes might yield more distant orthologs (Venkatesh et al., 2014). The final dataset contained 99 unique CD4 orthologs, ranging from teleost fish to human. These (putative) coding sequences were translated to amino acids and aligned using MAFFT (Katoh et al., 2002). For codon-based analyses, the aligned amino acid sequences were back translated to nucleotides to maintain codons. The multiple sequence alignments were further processed to remove all insertions (indels) relative to the mouse CD4 sequence (NM_013488.3) to maintain consistent numbering of sites. The 5' and 3' regions of the CD4 molecules were not consistently aligned due to different start codon usage or extensions of the ICD, respectively. The alignment was edited to start at the codon (AAG) coding for K48 within mouse CD4. The alignment that includes all 99 CD4 sequences ends at the last cysteine residue that makes up the CQC clasp. For the mammalian only dataset, the alignment terminates at the mouse CD4 stop codon.

FastTree was used to estimate maximum likelihood trees (Price et al., 2010). For amino acid-based trees, the Jones-Taylor-Thornton (JTT) model of evolution was selected, while the General Time-Reversible (GTR) model was used for nucleotide trees (Jones et al., 1992; Waddell and Steel, 1997). In all conditions, a discrete gamma model with 20 rate categories was used. A reduced representation tree (based on the amino acid alignment) is shown in **Figure 1A**. We used the same JTT model to estimate the marginal reconstructions of nodes indicated in **Figure 1A**. Phylogenetic trees and logo plots were visualized in Geneious and further edited using Adobe Illustrator. Ancestral sequences were estimated using GRASP (Foley et al., 2020).

Codon-based analysis of selection was performed using the hypothesis testing fixed effects likelihood (FEL) model as implemented within the phylogenies (HyPhy) package (version 2.5.14) (MP) (Kosakovsky Pond et al., 2020; Pond et al., 2005; Pond and Frost, 2005; Weaver et al., 2018). The back-translated codon-based alignments described above were used for these analyses. FEL uses likelihood ratio tests (LRTs) to assess a better fit of codons that allowed selection ($p < 0.1$). When calculating values for all CD4 orthologs included in the initial phylogenetic analysis we analyzed and identified the sequences within the mammalian clade as the foreground branches on which to test for evolutionary selection in order to maximize statistical power.

Covariation between protein residues was calculated using the MISTIC2 server. We calculated 4 different covariation methods (MIp, mFDCA, plmDCA, and gaussianDCA)(Colell et al., 2018). Protein conservation scores were calculated based on the protein alignment using the ConSurf Server (Ashkenazy et al., 2016; Ashkenazy et al., 2010). ConSurf conservation scores are normalized, so that the average score for all residues is zero, with a standard deviation of one. The lower the score, the more conserved the protein position. For the purpose of this study, residues were considered to covary if the MI was larger than 4 and both residues had a ConSurf conservation score lower than -0.5. Also, pairs with an MI larger than 8 were considered to covary if the conservation score was below -0.3. Using these criteria, we selected 0.5% of all possible pairs as recommended (Buslje et al., 2009; Colell et al., 2018).

Statistical analyses and associated raw data for the figures are available on Dryad (doi:10.5061/dryad.p8cz8w9vw).

Cell lines

58 α β ⁻ T cell hybridoma lines were generated from Kuhns Lab stocks of parental 58 α β ⁻ T cell hybridoma cells (obtained from Y.H. Chien at Stanford University) by retroviral transduction and maintained in culture by standard techniques as previously described (Glassman et al., 2018; Letourneur and Malissen, 1989). 58 α β ⁻ T cell hybridomas lack expression of endogenous TCR α and TCR β chains, are CD4 negative, make IL-2 in response to TCR signaling, and are variant of the DO-11.10.7 mouse T cell hybridoma (Balb/c T cell fused to BW5147 thymoma) (Letourneur and Malissen, 1989). We validate the cells lines by these characteristics as well as expression of H2-D^d to validate Balb/c origin. In brief, one day after transduction the cells were cultured in 5 μ g/ml puromycin (Invivogen) and 5 μ g/ml zeocin (ThermoFisher Scientific) in RPMI 1640 (Gibco) supplemented with 5% FBS (Atlanta Biologicals or Omega Scientific), Penicillin- Streptomycin- Glutamine (Cytiva), 10 μ g/ml Ciprofloxacin (Sigma), and 50 μ M beta-2- mercaptoethanol (ThermoFisher Scientific). The next day drug concentrations were increased to 10 μ g/ml puromycin (Invivogen) and 100 μ g/ml zeocin (ThermoFisher Scientific) in 10ml in a T25 flask. Aliquot of 1 $\times$ 10⁷ cells were frozen at days 5, 7, and 9. Cells were thawed from the day 5 freeze and cultured for 3 days in 10 μ g/ml puromycin and 100 μ g/ml zeocin, and maintained below 1 $\times$ 10⁶ cells/ml to use in the functional assays. Cells used in the functional assays were grown to 0.8 $\times$ 10⁶ cells/ml density and replicates of three functional assays were performed every other day. If cells exceeded 1 $\times$ 10⁶ cells/ml at any point in the process they were discarded as they lose reactivity at high cell densities and a new set of vials was thawed. Typically, two independent WT and mutant pairs were generated for any given mutant and tested for IL-2 to gain further confidence in a response phenotype. When cells lines are presented together in a graph, that indicates that the cell lines (WT and mutants sets) were generated at the same time from the same parental cell stock.

Given the number of mutant CD4 cell lines generated and handled in this study, the identity of the transduced CD4 gene was verified by PCR sequencing at the conclusion of three independent functional assays. 58 α β ⁻ T cell hybridomas were lysed using DirectPCR Tail Lysis Buffer (Viagen Biotech) with proteinase K (SIGMA) for 2 hours at 65°C. Cells were then heated at 95°C for ten minutes. Cell debris was pelleted, and the supernatants were saved. CD4 was amplified by PCR using Q5 DNA Hot Start Polymerase (New England BioLabs) in 0.2 μ M dNTP, 0.2 μ M primer concentration, Q5 Reaction buffer, and water. CD4 was amplified using the following primers:

5' primer: acggaattccgctcgagcgccaccatgggtgcgagccatctctctcttagg

3' primer: cttagcaagcttgctgactcaagatcttcattagatgagattatggctctctctgc

Product were purified using SpinSmart Nucleic Acid Purification Columns (Thomas Scientific) and sent to Eton Bioscience for sequencing with the following 5' CD4 primer: gtctctgaggagcagaag.

The I-E^k M12 cells used as antigen presenting cells (APCs) were previously reported (Glassman et al., 2018). M12 cells are a murine B cell lymphoma from Balb/c mice (H2-D^d validated) (Kim et al., 1979). Cells were cultured in RPMI 1640 (Gibco) supplemented with 5% FBS (Atlanta Biologicals or Omega Scientific), Pencillin- Streptomycin- Glutamine (Cytiva), 10 μ g/ml Ciprofloxacin (Sigma), 50 μ M beta-2- mercaptoethanol (ThermoFisher Scientific), and 5 μ g/mL Puromycin (Invivogen) and 50 μ g/ml Zeocin (ThermoFisher). The parental cells are maintained in Kuhns Lab stocks and were originally obtained from MM Davis stocks (Stanford University).

Parental 58 α β ⁻ T cell hybridoma and M12 cells are periodically treated with Plasmocyn and tested for mycoplasma contamination by PCR using primer sequences available from ATCC (5' primer sequence: TGCACCATCTGTCACTCTGTAACTC; 3' primer sequence:

GGGAGCAAACAGGATTAGATACCCT). All transduced cell lines used here were grown in ciprofloxacin and generated from parental cells that had tested negative for mycoplasma.

For retroviral production we used Phoenix-eco cells from the Nolan Lab (ATCC CRL-3214).

Antibodies

Antibodies	Vendor	Catalog number	RRID
anti-mouse CD4 eFlour 450, clone GK1.5	ThermoFisher Scientific	48-0041-82	AB_10718983
anti-mouse TCR α APC, clone RR8-1	ThermoFisher Scientific	17-5800-82	AB_19853170
anti-mouse CD3 ϵ PE-Cy7, clone 145-2C11	ThermoFisher Scientific	25-0031-82	AB_469572
anti-mouse IL-2, clone JES6-1A12	BioLegend	503702	AB_315292
biotin anti-mouse IL-2, clone JES6-5H4	BioLegend	503804	AB_315298
Streptavidin HRP	BioLegend	405210	
anti-mouse TCR β PE, clone KJ25	BD Pharmingen	553209	AB_394709
biotin anti-mouse CD4 (Clone RM4-4)	BioLegend	116010	AB_2561504
anti-mouse CD4 APC, clone GK1.5	BioLegend	100412	AB_312697
anti-mouse LCK PE, clone 3A5	Santa Cruz	sc-433	
Cholera Toxin Subunit B Alexa Fluor 488	ThermoFisher Scientific	C22841	
anti-mouse pCD3 ζ Alexa Fluor 647, clone K25-407.69	BD Phosflow	558489	AB_647152
anti-mouse pZAP70 APC, clone n3kbu5	ThermoFisher Scientific	17-9006-42	AB_2573268
anti-mouse pPLC γ 1 PE, clone A17025A	BioLegend	612404	AB_2801120

Flow cytometry

Cell surface expression of CD4 and TCR-CD3 complexes were measured by flow cytometry. In brief, cells were stained for 30 minutes at 4°C in FACS buffer (PBS, 2% FBS, and 0.02% sodium azide) using anti-CD4 (clone GK1.5, eFluor 450 conjugate, ThermoFisher Scientific), anti-TCR α (anti-V α 11, clone RR8-1, APC conjugate, ThermoFisher Scientific), anti-CD3 χ (145-2C11, ThermoFisher Scientific) and GFP was detected as a measure of the TCR β -GFP subunit. Analysis was performed on a Canto II or LSRII (BD Biosciences) at the Flow Cytometry Shared Resource at the University of Arizona. Flow cytometry data were analyzed with FlowJo Version 9 software (Becton, Dickinson & Company).

Functional assays

IL-2 production was measured to quantify pMHCII responses. 5×10^4 transduced 58 $\alpha\beta^-$ T cell hybridomas were cocultured with 1×10^5 transduced I-E $^{k+}$ M12 cells in triplicate in a 96 well round bottom plate in RPMI with 5% FBS (Omega Scientific), Pen-Strep+L- Glutamine (Cytiva), 10ng/ml ciprofloxacin (Sigma), and 50 μ M beta-2-mercaptoethanol (Fisher) in the presence of titrating amounts of MCC 88-103 peptide (purchased from 21st Century Biochemicals at >95% purity) starting at 30 μ M MCC and a 1:3 titration (Glassman et al., 2018 [DOI](#)). For experiments with APCs

expressing tethered pMHCII, 5×10^4 58 α β ⁻ T cell hybridomas were cultured with 1×10^5 MCC:I-E^{k+} or T102S:I-E^{k+} M12 cells in triplicate in a 96 well round bottom plate using the same culture conditions as above. The supernatants were collected and assayed for IL-2 concentration by ELISA after 16 hours of co-culture at 37°C. Anti-mouse IL-2 (clone JES6-1A12, BioLegend) antibody was used to capture IL-2 from the supernatants, and biotin anti-mouse IL-2 (clone JES6-5H4, BioLegend) antibody was used as the secondary antibody. Streptavidin- HRP (BioLegend) and TMB substrate (BioLegend) were also used.

To assess engagement-induced endocytosis, CD4 surface levels were measured by flow cytometry 16hrs after coculture with APCs and peptide as described above for IL-2 quantification. 96 well plates containing cells were washed with ice cold FACS buffer (PBS, 2%FBS, 0.02% sodium azide), transferred to ice, and Fc receptors were blocked with Fc block mAb clone 2.4G2 for 15 minutes at 4°C prior to surface staining for 30 minutes at 4°C with anti-CD4 (clone GK1.5 EF450, Invitrogen) and anti-V β 3 TCR clone (clone KJ25, BD Pharmingen) antibodies. Cells were washed with FACS buffer prior to analysis on a LSRII (BD Biosciences) at the Flow Cytometry Shared Resource at the University of Arizona. Flow cytometry data were analyzed with FlowJo Version 10 software (Becton, Dickinson & Company). The average of the geometric mean of the TCR or CD4 signal was taken for the triplicate of the post 58 α β ⁻ cells cocultured with M12 I-E^{k+} cells at 0 μ M MCC concentration. Each value of the raw gMFI of TCR or CD4 for cells cultured at 10 μ M MCC was subtracted from the average gMFI at 0 μ M. The values show the change of gMFI from 0 μ M to 10 μ M.

Sucrose gradient analysis

Membrane fractionation by sucrose gradient was performed similarly to previously described methods (Hur et al., 2003 [DOI](#); Parrish et al., 2016 [DOI](#)). For cell lysis, 6×10^7 58 α β ⁻ T cell hybridomas were harvested and washed 2x using TNE buffer (25mM Tris, 150mM NaCl, 5mM EDTA). Cells were lysed on ice in 1% Triton-X detergent in TNE in a total volume of 1mL for 10 minutes and then dounce homogenized 10x. The homogenized lysate was transferred to 14 x 95mm Ultraclear Ultra Centrifuge tubes (Beckman). The dounce homogenizer was rinsed with 1.6mL of the 1% lysis buffer, which was then added to the Ultracentrifuge tube. 2.5mL of 80% sucrose was added to the centrifuge tube with lysate and mixed well. Gently, 5mL of 30% sucrose was added to the centrifuge tubes, creating a 30% sucrose layer above the ~40% sucrose/lysate mixture. Then, 3mL of 5% sucrose was added gently to the centrifuge tube, creating another layer. The centrifuge tubes were spun 18 hours at 4°C in a SW40Ti rotor at 36,000rpm.

Analysis of membrane fractions was performed via flow-based fluorophore-linked immunosorbent assay (FFLISA) as previously described (Parrish et al., 2016 [DOI](#)). In brief, 88 μ L of Streptavidin Microspheres 6.0 μ m (Polysciences) were coated overnight at 4°C with 8 μ g of biotinylated anti-CD4 antibody (clone RM4-4, BioLegend). Prior to immunoprecipitation (IP), beads were washed with 10mL of FACS buffer (1x PBS, 2%FBS, 0.02% sodium azide) and resuspended in 3.5mL of 0.1% Triton-X 100 lysis wash buffer in TNE. For each cell line lysed, 10 FACS tubes were prepared with 50 μ L of the washed RM4-4 coated beads. Upon completion of the spin, 500 μ L was carefully taken off the top of the centrifuge tubes and discarded. Following this, 1mL was extracted from the top of the tube, carefully as to not disrupt the gradient, and added to a FACS tube with coated beads and capped. This was repeated for 10 individual fractions in separate FACS tubes. Following the extraction, lysates were incubated with the beads for 90 minutes, inverting the tubes to mix every 15 minutes.

Following the immunoprecipitation, FACS tubes were washed 3x using 0.1% Triton-X lysis wash buffer in TNE. Tubes were then stained using 1 μ L anti-CD4 (APC conjugate; clone GK1.5, BioLegend), 1.5 μ L anti-LCK (PE conjugate, clone 3A5, Santa Cruz Biotechnology), and 1 μ L CTxB (AF 488 conjugate, ThermoFisher Scientific, resuspended as per manufacturer's instructions) for 45 minutes at 4°C. Following the stain, tubes were washed using 0.1% Triton-X lysis wash buffer in

TNE. Analysis of beads were performed on a LSRII (BD Biosciences) at the Flow Cytometry Shared Resource at the University of Arizona. 10^4 events were collected per sample. Flow cytometry data were analyzed with FlowJo Version 10 software (Becton, Dickinson & Company).

For FFLISA analysis, raw gMFI values for fraction 1 were subtracted from the rest of the fractions to account for background, such that the gMFI of fraction 1 is 0. To normalize the data, the percentage of CD4 within any given fraction (fx) relative to the total CD4 gMFI (CD4 signal % of total) was calculated by dividing the gMFI signal in a given fraction (fx) by the sum of the total CD4 gMFI signal [sum(f1:f10)CD4 gMFI] and multiplying by 100 [e.g. fx % of total = fx CD4 gMFI / Sum(f1:f10)CD4 gMFI x 100]. To normalize the CTxB and LCK signal in any given fraction relative to the CD4 signal in that same fraction (CTxB or LCK normalized to CD4) the gMFI of CTxB or LCK in fx was divided by the CD4 gMFI of that fx and then multiplied by the percentage of CD4 within fx (e.g. Normalized fx LCK = fx LCK gMFI / fx CD4 gMFI x fx CD4% of total CD4 gMFI). Area under the curve (AUC) analysis was performed with GraphPad Prism 9 for fractions 1-6 to determine the AUC for the detergent resistant membrane (DRM) domains due to their floating phenotypes, and for fractions 6-10 to determine the AUC for the detergent soluble membrane (DSM) domains.

Intracellular signaling analysis

M12 cells expressing Hb:I-E^k (null) or MCC:I-E^k (cognate) tethered pMHCII complexes were labeled with Tag-it Violet according to the manufacturer's instructions (BioLegend). M12 cells and 58αβ⁻ cells were then chilled on ice for 30 minutes, 5×10^5 of each cell type were mixed together in 1.5ml snap cap tubes, and the cells were pelleted at 2000 rpm for 30 second at 4°C to force interactions. The supernatant was removed and the tubes were transferred to a 37°C water bath for 2 minutes to enable signaling. Fixation Buffer (BioLegend Inc) was then added for 15 minutes at 37°C. Cells were washed twice with FACs buffer, pelleted at 350xg for 5 minutes at room temperature, resuspended in 1mL True-Phos Perm Buffer (BioLegend Inc), and incubated at -20°C for 16 hours.

Cells were blocked with anti-mouse FcRII mAb clone 2.4G2 hybridoma supernatants (ATCC) for 30 minutes, pelleted, and stained on ice for 60 minutes with anti-pCD3ζ (clone K25-407.69, Alexa Fluor 647 conjugate, BD Biosciences) in one sample tube, or with anti-pZAP70 (clone n3kobu5, APC conjugate, Invitrogen) and anti- pPLCy1 (clone A17025A, PE conjugate, BioLegend) in a separate sample tube at the vendor-recommended concentrations. Finally, cells were washed 2x with FACs buffer at 1000xg for 5 minutes at room temperature and analyzed on a Canto II (BD Biosciences) at the Flow Cytometry Shared Resource at the University of Arizona or on a BD Fortessa. 1×10^4 58αβ⁻ cell:M12 cell couples were collected per sample.

Flow cytometry data were analyzed with FlowJo Verison 10 software (Becton, Dickinson, & Company) by gating on 58αβ⁻ and M12 cell couples, as described previously (Glassman et al., 2018 [10.1016/j.jbc.2018.01.010](https://doi.org/10.1016/j.jbc.2018.01.010)). Histograms of the pCD3ζ, pZAP70, or pPLCy1 intensity for the gated population were then generated and data expressing the gated populations as numbers of cells within intensity bins was exported from FlowJo into Microsoft Excel where the number of cells for each bin intensity value for MCC:I-E^k stimulated cells was subtracted from Hb:I-E^k stimulated cells on a bin-by-bin basis. This allowed us to enumerate the intensity differences per bin upon stimulation with the agonist pMHCII over background. Mean intensity and standard error of the mean (SEM) were calculated based on the background subtracted (MCC:I-E^k-Hb:I-E^k) data. The data was then transferred to Prism 9 where we performed smoothing analysis with 500 nearest neighbors to smooth the line profile for graphing purposes. Those intensity bins with positive values were considered to contain cells that had responded to the MCC:I-E^k stimuli above background.

Statistical Analysis

Statistical analyses of sucrose gradient and functional assays were performed with GraphPad Prism 9 software as indicated in the figure legends. For each functional assay (IL-2 production and CD4 endocytosis), each individual experiment (biological replicate) was performed with triplicate

analysis (technical replicates) and each experiment was repeated three times (3 biological replicates). For sucrose gradient analysis, 10^4 beads were collected by flow cytometry in each experiment (technical replicates) and each experiment was performed three times (biological replicates).

Three biological replicates were chosen for each analysis as per convention, and no power calculations were determined. One-Way ANOVA was performed with a Dunnett's posttest when all mutants tested in an experiment were compared to a control sample (e.g. WT). Sidak's posttest were applied when comparing between two specific samples. These posttests were chosen based on Prism recommendations. Student's unpaired t-tests (two-tailed) were performed when comparing WT and LL mutant samples only for phosphorylation analysis.

Constructs

The sequences of 5cc7 α , 5cc7 β G, CD4 WT and CD4 T1 constructs were previously described (Lee et al., 2022 [DOI](#)). Full-length CD3 δ , ϵ , γ , and ζ were encoded on a poly- cistronic construct as previously described (Holst et al., 2008 [DOI](#); Kuhns and Davis, 2007 [DOI](#)).

The following CD4-T1 constructs: T1-TMD, T1-Palm (2C), T1-Palm (3C), T1-TP (2C), and T1-TP(3C) were cloned by conventional molecular biology techniques, including PCR-based mutagenesis where needed, into pUC18 via 5' EcoRI and 3' HindIII. After sequence verification they were subcloned into pP2 puromycin-resistance MSCV vector (MCS-IRES-puro) via 5' XhoI and 3' BglII, midi-prepped (Qiagen), and sequence verified. Mutated codons are shown in bold uppercase letters, while the motif under interrogation is shown in bold and underlined (any non-mutated codons within the motif are bold lowercase letters and underlined). The sequences for these constructs are as follows:

T1 tmd

```
acggaattccgctcgagcgccaccatggtgcgagccatctcttaggcgcttgctgctgctgctgctgcagctgtcacaac
tcctagctgtcactcaagggaagacgctggtgctgggaaggaagggaatcagcagaactccctgcgagagtccc
agaagaagatcacagtcttcacctggaagtctctgaccagaggaagattctggggcagcatggcaagggttattaatta
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aaacttaagatggaagactctcagacttatactgtgagctggagaacaggaaaggaggtggagttgtgggtgttcaa
agtgaccttcagtcgggtaccagcctgttgcaaggcgagagcctgacacctgagtagcaacttaaggctcttaa
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agaggtgtccgtacaaaagtcaccaaagacccaagctcagctgaaggaaacgctccactcaccctcaagataacc
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ctgaagcaggagaaccaggagggcagggtctctgaggagcagaaagtagttcaagtggtggccctgagacagggtct
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cagtggtcctggtgctgctggtggtGtGtcttcCTCtttctgggttcttgggctctgcatcctctgctgtcaggtgcc
ggtgataaagatcttgagtcgacaagcttgctag
```

T1-Palm (2C)

```
acggaattccgctcgagcgccaccatggtgcgagccatctcttaggcgcttgctgctgctgctgctgcagctgtcacaac
tcctagctgtcactcaagggaagacgctggtgctgggaaggaagggaatcagcagaactccctgcgagagtccc
agaagaagatcacagtcttcacctggaagtctctgaccagaggaagattctggggcagcatggcaagggttattaatta
gaggaggttcgcttcgagtttgatcgtttgattccaaaaaggggcatgggagaaaggatcgtttctctcatcatcaat
aaacttaagatggaagactctcagacttatactgtgagctggagaacaggaaaggaggtggagttgtgggtgttcaa
agtgaccttcagtcgggtaccagcctgttgcaaggcgagagcctgacacctgagtagcaacttaaggctcttaa
```

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ttcagagcacagctatcacggcctataagagtgaggggagagtcagcggagttctcctcccactcaactttgcagaggaa
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cggtgataaagatcttgagtcgacaagcttgctag

T1-Palm (3C)

acggaattccgctcgagcgccacattggtcgagaccatctctttaggcgcttgctgctgctgctgagctgcagctgcacaaac
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cagtgttcctggcttgctgctgggtggctccttcggcttcttgggttcttgggctctgcatctc**TCCTCTgtcaggTC**
Ccggtgataaagatcttgagtcgacaagcttgtag

T1 tp 2 c

acggaattccgctcgagcgccaccatgggtgcgagccatctctttaggcgcttgcgtgctgctgctgcagctgcacaa
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ccccttgacagagtgcacaacacaaaaagggtaaagtgtcagtggttccaaagtctctccatgtccaacctagggttca
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cagtgttctcgcttcgctgctggtGTGtcttcCTCttctgggtttccttgggctctgcatcctctgcTCTgtcaggT
CCcgggtgataaagatcttgagtcgacaagcttgctag

T1 tp 3 c

acggaattccgctcgagcgccaccatggtgcgagccatctctttaggcgcttgctgctgctgctgctgagctgtcacaac
tcctagctgtcactcaagggaagacgctggtgctggggaagggaaggggaatcagcagaactgccctgcgagagttccc
agaagaagatcacagtcttcacctggaagtctctgaccagaggaagattctggggcagcatggcgaagggtgattaata
gaggagggttcgcttcgagtttgatcgttttgattccaaaaaaggggcagtgaggagaaggatcgttctctcatcatcaat
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agtaccttcagtcgggtaccagcctgttgcaagggcagagcctgacccctgaccttgatagcaacttaaggctctaa
ccccttgacagagtgcacaacaaaaagggtaaagtgtcagtggttccaaagtctctccatgtccaaactaagggttca
ggacagcgactctggaactgcaccgtgacccctggaccagaaaaaagaactggttcggcatgacactctcagtgctgggt
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gtggcagtgctactagtgaaagtgataaggtcaagatggactccaggtatccaggttttatccagagggtgaaccaga
cagtggttcgtgcttgctgctggt**GTGtcctcCTC**tttctgggttcttgggtctgcacatcct**TCCTCTgtcagg**
TCCcggtgataaagatcttgagtcgacaagctgctag

The gene of T1^{Δbind} was made by cloning the in frame extracellular domain of CD4 that encodes the Δbind extracellular domain (Parrish et al., 2015 [DOI](#)) flanked by XhoI and BamHI into the T1 gene. The gene was sequence verified in pUC 18, subcloned into pP2 puromycin-resistance MSCV vector (MCS-IRES-puro) via 5' XhoI and 3' BglII, midi- prepped, and sequence verified. The sequence for the construct is as follows with mutated codons shown in bold uppercase letters, while the motif under interrogation is shown in bold and underlined (any non-mutated codons within the motif are bold lowercase letters and underlined):

T1^{Δbind}

acggaattccgctcgagcgccaccatggtgcgagccatctctttaggcgcttgctgctgctgctgctgagctgtcacaac
tcctagctgtcactcaagggaagacgctggtgctggggaagggaaggggaatcagcagaactgccctgcgagagttccc
agaagaagatcacagtcttcacctggaagtctctgaccagaggaagattctggggcagcatggc**GATggtGATTC**
AGATAGCggaggttcgcttcgagtttgatcgttttgattccaaaaaaggggcagtgaggagaaggatcgttctctc
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ggtgttcaagtgaccttcagtcgggtaccagcctgttgcaagggcagagcctgacccctgaccttgatagcaactctaa
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ggtgccggtgataaagatcttgagtcgacaagctgctag

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Editors

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Reviewer #1 (Public Review):

Summary:

This study by Lee et al. is a direct follow-up on their previous study that described an evolutionary conservancy among placental mammals of two motifs (a transmembrane motif and a juxtamembrane palmitoylation site) in CD4, an antigen co-receptor, and showed their relevance for T-cell antigen signaling. In this study, they describe the contribution of these two motifs to the CD4-mediated antigen signaling in the absence of CD4-LCK binding. Their approach was the comparison of antigen-induced proximal TCR signaling and distal IL-2 production in 58^{-/-} T-cell hybridoma expressing exogenous truncated version of CD4 (without the interaction with LCK), called T1 and T1 version with the mutations in either or both of the conserved motifs. They show that the T1 CD4 can support signaling to extend similar to WT CD4, but the mutation of the conserved motifs substantially reduced the signaling. The authors conclude that the role of these motifs is independent of the LCK-binding.

Strengths:

The authors convincingly show that CD4 is capable of contributing to TCR signaling in a manner independent of LCK, but dependent on the two studied motifs in CD4.

Weaknesses:

- (1) Experiments in primary T cells are required to estimate the relative contribution of LCK-dependent and LCK-independent mechanisms of CD4 signaling.
- (2) The mechanistic explanation (beyond the independence of LCK binding) of the role of these motifs is unclear at the moment.

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Reviewer #2 (Public Review):

Summary:

The paper by Kuhn and colleagues follows upon a 2022 eLife paper in which they identified residues in CD4 constrained by evolutionary purifying selection in placental mammals, and then performed functional analyses of these conserved sequences. They showed that sequences distinct from the CXC "clamp" involved in recruitment of Lck have critical roles in TCR signaling, and these include a glycine-rich motif in the transmembrane (TM) domain and the cys-containing juxtamembrane (JM) motif that undergoes palmitoylation, both of which promote TCR signaling, and a cytoplasmic domain helical motif, also involved in Lck binding, that constrains signaling. Mutations in the transmembrane and juxtamembrane sequences led to reduced proximal signaling and IL-2 production in a hybridoma's response to antigen presentation, despite retention of abundant CD4 association with Lck in the detergent-soluble membrane fraction, presumably mislocalized outside of lipid rafts and distal to the TCR. A major conclusion of that study was that CD4 sequences required for Lck association,

including the CXC "clasp" motif, are not as consequential for CD4 co-receptor function in TCR signaling as the conserved TM and JM motifs. However, the experiments did not determine whether the functions of the TM and JM motifs are dependent on the Lck-binding properties of CD4 - the mutations in those motifs could result in free Lck redistributing to associate with CD4 in signaling-incompetent membrane domains or could function independently of CD4-Lck association. The current study addresses this specific question.

Using the same model system as in the earlier eLife paper (the entire methods section is a citation to the earlier paper), the authors show that truncation of the Lck-binding intracellular domain resulted in a moderate reduction in IL-2 response, as previously shown, but there was no apparent effect on proximal phosphorylation events (CD3z, Lck, ZAP70, PLCg1). They then evaluated a series of TM and JM motif mutations in the context of the truncated Lck-nonbinding molecule and showed that these had substantially impaired co-receptor function in the IL-2 assay and reduced proximal signaling. The proximal signaling could be observed at high ligand density even with a MHC non-binding mutation in CD4, although there was still impaired IL-2 production. This result additionally illustrates that phosphorylation of the proximal signaling molecules is not sufficient to activate IL-2 expression in the context of antigen presentation.

Strengths:

The strength of the paper is the further clear demonstration that the classical model of CD4 co-receptor function (MHCII-binding CD4 bringing Lck to the TCR complex, for phosphorylation of the CD3 chain ITAMs and of the ZAP70 kinase) is not sufficient to explain TCR activation. The data, combined with the earlier eLife paper, further implicate the gly-rich TM sequence and the palmitoylation targets in the JM region as having critical roles in productive co-receptor-dependent TCR activation.

Weaknesses:

The major weakness of the paper is the lack of mechanistic insight into how the TM and JM motifs function. The new results are largely incremental in light of the earlier paper from this group as well as other literature, cited by the authors, that implicates "free" Lck, not associated with co-receptors, as having the major role in TCR activation. It is clear that the two motifs are important for CD4 function at low pMHCII ligand density. The proposal that they modulate interactions of TCR complex with cholesterol or other membrane lipids is an interesting one, and it would be worth further exploring by employing approaches that alter membrane lipid composition. The JM sequence presumably dictates localization within the membrane, by way of palmitoylation, which may be critical to regulate avidity of the TCR:CD4 complex for pMHCII or TCR complex allosteric effects that influence the activation threshold. Experiments that explore the basis of the mutant phenotype could substantially enhance the impact of this study.

Additional comments:

- Is the "IL-2 sensitivity" measurement for the T1-TP (3C) meaningful (Table 3)? It is showing only a moderate reduction compared to T1 control, while TP (2C) or just the 3C palmitoylation mutations essentially eliminate response.
- It is unclear how the pairs of control and mutant cells connected by lines in the figures are related. They are presumably cells from distinct biological experiments, with technical replicates for each, but are they paired because they were derived at the same time with different constructs? This should be explained in this paper, not in a reference.

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Author Response

The following is the authors' response to the previous reviews.

Reviewer #1 (Public Review):

Summary:

This study by Lee et al. is a direct follow-up on their previous study that described an evolutionary conservancy among placental mammals of two moBfs (a transmembrane moBf and a juxtamembrane palmitoylation site) in CD4, an anBgen co-receptor, and showed their relevance for T-cell anBgen signaling. In this study, they describe the contribution of these two moBfs to the CD4-mediated anBgen signaling in the absence of CD4-LCK binding. Their approach was the comparison of anBgen-induced proximal TCR signaling and distal IL-2 production in 58-/- T-cell hybridoma expressing exogenous truncated version of CD4 (without the interaction with LCK), called T1 with T1 version with the mutations in either or both of the conserved moBfs. They show that the T1 CD4 can support signaling to the extent similar to WT CD4, but the mutation of the conserved moBfs substantially reduced the signaling. The authors conclude that the role of these moBfs is independent of the LCK-binding.

Strengths:

The authors convincingly show that T1 CD4, lacking the interaction with LCK supports the TCR signaling and also that the two studied moBfs have a significant contribution to it.

Weaknesses:

The study has several weaknesses.

(1) The whole study is based on a single experimental system, genetically modified 58-/- hybridoma. It is unclear at this moment, how the molecular moBfs studied here contribute to the signaling in a real T cell. The evolutionary conservancy suggests that these moBfs are important for T cell biology. However, the LCK-binding moBf is conserved as well (perhaps even more) and it plays a very minor role in their model. Without verifying their results in primary cells, the quantitative, but even qualitative, importance of these moBfs for T-cell signaling and biology is unclear. Although the authors discuss this issue in the Discussion, it should be noted in all important parts of the manuscript, where conclusions are made (abstract, end of introduction, perhaps also in the title) that the results are coming from the hybridoma cells.

We appreciate the Reviewer's thoughtful comments and suggestions. We now state in the abstract and introduction that wet-lab experiments were performed with T cell hybridomas. We have also better highlighted work from Killeen and LiXman (PMID: 8355789) wherein they showed that C-terminally truncated CD4, which lacked the moBfs that mediate CD4-Lck interactions, can drive CD4+ T cell development, proliferation, and T-helper function because we now provide mechanistic data to help explain those in vivo results. Also, as noted by the reviewer, we discuss how the sum of our data provides justification for the investment in and use of mouse models to interrogate how the functionally important residues/moBfs identified and studied here influence T cell biology.

We will take the opportunity to reiterate here that, while the study is based on a well characterized, albeit single, wet-lab experimental system, the whole study is based on two lines of investigation. The other approach was a systems biology computational approach that analyzes data from real-world experiments in a variety of jawed vertebrate species over evolution. Specifically, we used a computational reconstruction of the evolutionary history of

CD4 by performing multiple analyses of CD4 from 99 jawed vertebrates spanning ~435 million years of evolution. This analysis allowed us to identify residues, and networks of evolutionarily coupled residues, that are predicted to be functionally important in vivo. Like other systems biology approaches, this allowed us to look at the larger picture by evaluating data points that have emerged from constant testing and adjustments of CD4 function in vivo through selection on an evolutionary timescale in more jawed vertebrate species, and under more real-world conditions, than can be tested in the laboratory. Our structure-function analysis provided a second, wet-lab reductionist experimental system to cross-validate that the residues identified by our evolutionary analysis are functionally significant. This experimental validation is critical and elevates the relevance of our studies above ad hoc observations. Our work also provides mechanistic insights for why the residues studied here are functionally significant (i.e., key determinants of pMHCII-specific signaling initiation). In short, using both systems allowed us to cross-validate the functional significance of the residues within the GGXXG and (C/F)CV+C motifs studied here by two independent methods.

(2) Many of the experiments lack the negative control. I believe that two types of negative controls should be included in all experiments. First, hybridoma cells without CD4 (or with CD4 mutant unable to bind MHCII). Second, no peptide control, i.e., activation of the hybridoma cells with the APC not loaded with the cognate peptide. These controls are required to distinguish the basal levels of phosphorylation and CD4-independent antigen-induced phosphorylation to quantify, what is the contribution of the particular motifs to the CD4-mediated support. Although these controls are included in some of the experiments, they are missing in other ones. The binding mutant appears in some FC results as a horizontal bar (without any error bar/variability), showing that CD4 does not give a huge advantage in these readouts. Why don't the authors show no peptide controls here as well? Why the primary FC data (histograms) are not shown? Why neither of these two controls is shown for the % of responders plots? Although the IL-2 production is a very robust and convincing readout, the phosphoflow is much less sensitive. It seems that the signaling is elevated only marginally. Without the mentioned controls and showing the raw data, the precise interpretation is not possible.

These comments, and those in point #3, concern our flow cytometry-based analysis of early intracellular signaling events where we asked: how do the motifs under investigation impact phosphorylation of CD3z, ZAP-70, and PLCγ1 in response to agonist pMHCII? Thank you for pointing out areas of confusion regarding these analyses. We will try to clarify here and have worked to clarify the text.

Our approach was to mutate conserved residues within the motifs that our evolutionary analysis predicted to be functionally significant, compare the performance of the mutants to that of controls bearing WT motifs, and then infer the function of the motifs based on the differential phenotype of the mutants relative to their controls. In most cases, the C-terminally truncated CD4-T1 mutant served as the appropriate CD4 control backbone against which to evaluate the phenotypes of the GGXXG and (C/F)CV+C motif mutants. This is a conventional structure-function strategy.

All experiments included APCs expressing null pMHCII (Hb:I-Ek) as negative controls. These were a necessary component of the data analysis, explained further below, which involved background subtraction of the signal from control or mutant T cell hybridomas bound to these negative control APCs from those bound to the agonist pMHCII (MCC:I-Ek). Doing so allowed us to establish a true signal over background for calculating percent responders and signaling intensity. These negative controls served the same purpose of APCs expressing I-Ek not loaded with cognate peptide requested by the reviewer. It is important to note that we previously published that TCR-CD3-pMHCII interactions reciprocally increase CD4-pMHCII dwell time, and vice versa, such that dwell times of the 5c.c7 TCR and CD4 to the null Hb:I-Ek are both basal in this system relative to antagonist, weak agonist, and agonist pMHCII (PMID

29386113). A recent study using different techniques also concluded that TCR-CD3 and CD4 cooperatively enhance signaling to pMHCII (PMID 36396644). The use of the null pMHCII, Hb:I-Ek, in each experiment thus serves as a well-characterized negative control for both TCR and CD4 engagement in this experimental system with regards to assembly of the TCR-CD3 and CD4 around pMHCII to drive signaling. In our view, it is the most important negative control for interpreting our results, and it is present in each experiment. In Fig 1B and related supplemental figures we compare the Cterminally truncated CD4-T1 mutant to the full-length WT CD4 to evaluate the contributions of the intracellular domains to early signaling events. We found no significant differences for pCD3z, pZAP-70, and pPLCg1 levels demonstrating that, in our system, CD4 WT and T1 are statistically indistinguishable.

In Fig 1C we asked: what is the contribution of CD4-pMHCII interactions made by CD4 T1, which lacks the intracellular domain, using our CD4 T1Dbind mutant. Fig 2C and Table 3 show that pCD3z levels for T1Dbind were ~54% of T1, meaning that CD4 binding to pMHCII roughly doubles pCD3z levels (even without the intracellular domain). We also showed that the percent of responders were not different between the CD4 T1 and T1Dbind mutant in Fig 2C. The impact on ZAP-70 and PLCg1 are shown in Figure 2—figure supplement 4. These differences, including the magnitude of the decrease, were observed reproducibly ($p < 0.001$) in three independently generated sets of lines. We believe that this analysis satisfies the request by the reviewer for an analysis of the contributions of CD4 binding to pMHCII. We did not include this as a negative control in experiments evaluating the contributions of the GGXXG and (C/F)CV+C motifs to CD4 T1 signaling because the question being asked in those experiments was how do the motifs impact signaling in the absence of the intracellular domain (i.e., within the CD4 T1 backbone, making CD4 T1 the proper comparator for the question we were asking). We showed the average normalized intensity for the T1Dbind mutant, relative to T1, for this lower bound of signaling mediated by TCR-CD3-only as a dotted line in those figures to provide a reference point for the readers to evaluate and put into perspective how the mutants we generated impacted the overall contribution of CD4 to these early signaling events. The T1Dbind mutants were not always measured in the same experiment at the same time with other mutants, because the cell lines used were not always made at the same time, so we did not think it appropriate to graph the results together.

We do not know how to interpret the comment “Although the IL-2 production is a very robust and convincing readout, the phosphoflow is much less sensitive. It seems that the signaling is elevated only marginally.” We will offer our perspective that we do not know how to equate the sensitivity of the phosphoflow to the IL-2. Because the IL-2 is a signaling output, it results from signaling amplification from the membrane to the nucleus. If CD3z phosphorylation is the initiating event for a signaling cascade that leads to IL-2 gene transcription and transduction, as is widely believed, our data strongly suggests that the ~2-fold difference in pCD3z levels between CD4 T1 and T1Dbind (Fig 2C/Table 3 data) contributes to the difference between no IL-2 output for T1Dbind and IL-2 output by T1 in this experimental system. Because CD4 WT and T1 have significantly different levels of IL-2 output, but show no significant differences in pCD3z, pZAP-70, or pPLCg1 levels, there are likely to be other differences we did not measure via other pathways that intersect at the nucleus. At many levels, biology works on gradients such that small differences can bias a system in one direction or another. The kinetic discrimination model (PMID 8643643), which is thought to be a reasonable description of the relationship between pMHC engagement and signaling outcomes, suggests that very small differences in molecular interactions at the earliest stages of a response can lead to big differences in signaling outcome. We therefore have no basis at this juncture to think that ~2-fold differences in pCD3z levels could not account for bigger differences in signaling output such as IL-2.

(3) The processing of the data is not clear. Some of the figures seem to be overprocessed. For instance, I am not sure what “Normalized % responders of pCD3zeta” means (e.g., Fig. 1C and elsewhere)? Why do not the authors show the actual % of pCD3zeta+ cells

including the gaBng strategy? Why do the authors subtract the two histograms in Fig. 2-Fig.S3? It is very unusual.

We did develop and implement a novel strategy for measuring the impact of our mutaBons on CD3z, ZAP-70, and PLCg1 phosphorylaBon. This was explained in more detail in our prior study. The instrucBons to authors indicated that we should not repeat methods in the current manuscript. However, we will go through the approach here, and address why we did not show primary FC histograms for all experiments from above. First, we think that a brief explanaBon as to what moBvated us to develop our approach will add to a beXer understanding:

- (1) For experimental and staBsBcal rigor, our goal was to perform both experimental and biological replicates by measuring and comparing the average of at least three independently generated sets of paired WT/T1 control Vs. mutant cells lines generated at different Bmes to determine the staBsBcal significance of the difference, if any, between averages of the control and mutant lines.
- (2) Our quesBons necessitated that we measure signals generated naturally by the cooperaBve engagement of cognate pMHCII by TCR-CD3 and CD4 on APCs, rather than through aCD3/aCD4 crosslinking.
- (3) We chose to use flow cytometry rather than bulk cell analysis by Western Bloung to analyze signaling occurring in cells that were engaged to the agonist APC in order to avoid diluBon of that signal by cells that are not engaged to APCs and not signaling. 4. For each experiment, we wanted to subtract background signals from cells bound to APCs expressing a null pMHCII (Hb:I-Ek) from signals generated by cells bound to APCs expressing agonist pMHCII (MCC:I-Ek). Doing so allowed us to idenBfy cells that are signaling (responders) to agonist over null pMHCII. The goal here was to quanBtate the level of signaling in an objecBve manner with a method that can be applied to all samples uniformly rather than seung a flow cytometry gate on posiBve cells (e.g. pCD3z) because gaBng is subjecBve and can vary from experiment to experiment. To put that another way, as detailed below, we used our subtracBon method to idenBfy signaling responders rather than seung a signaling gate on the posiBve populaBon.

Regarding gaBng schemes, controls, and data processing:

Figure 2—figure supplement 3 of the current study and Figure 6—figure supplement 1 of our prior study are designed to walk the reader through our experimental design, gaBng, data processing and thinking. Here we will provide a detailed explanaBon to complement the figure legend as well as the methods provided in our prior manuscript (see pt #4 below).

We will refer to Figure 2—figure supplement 3 here:

Panel A. The dot plots show our approach to idenBfying 5c.c7+ CD4+ 58a-b- T cell hybridomas (yaxis, GFP posiBve) coupled to M12 cells (x-axis, TagIt Violet) expressing the null pMHCII Hb:I-Ek (lev) or agonist pMHCII MCC:I-Ek (right). The gaBng shows the frequency of GFP+ T cell hybridomas that are bound to TagIt violet posiBve APCs (i.e., cell couples). The histogram on the right then shows the staining intensity for pCD3z on the x-axis for the 10,000 coupled events collected wherein the APCs express the null pMHCII (filled cyan) or the agonist pMHCII (black line).

Panel B. The data presented here is the same as in Panel A, but for CD4 T1 cells.

Panel C. The data presented here walks through how we idenBfy 5c.c7+ CD4+ 58a-b- T cell hybridomas responding (i.e., signaling) to agonist pMHCII, as well as the mean signaling intensity of the responding populaBon, in a gaBng-independent manner aver background subtracBon. For the lev graph, we exported the data for the histograms shown in Panel A

from FlowJo 10 software and plotted them here using Prism 9 as smoothed lines (500 nearest neighbors). The cyan line is therefore a replicate of the flow cytometry histogram shown in Panel A for pCD3z intensity from 5c.c7+ CD4+ 58a-b- T cell hybridomas coupled to M12 cells expressing the null pMHCII (Hb:I-Ek), while the black histogram is a replicate of the pCD3z intensity for 5c.c7+ CD4+ 58a-b- T cell hybridomas coupled to M12 cells expressing the agonist pMHCII (MCC:I-Ek). Next, to identify the responding population in a gaBng-independent manner, we used Excel to subtract the pCD3z intensity for the null pMHCII (cyan) negative control population on a bin-by-bin basis from the pCD3z intensity for the agonist pMHCII (black) responding population. We then transferred the background subtracted values to Prism 9 for smoothing and plotting (grey line: MCC:I-Ek minus Hb:I-Ek). The middle graph shows the same data processing for the data from Panel B for the CD4 T1 cells. Please note that the background subtracted grey line has negative values and positive values. The negative values represent intensity bins where signaling in response to agonist pMHCII leads to fewer cells per bin than in the null pMHCII population that is not signaling, while the positive values represent bins of intensity where signaling cells outnumber non-signaling cells. The right graph in this panel shows the populations average background subtraction for intensity bins that had more cells with pCD3z signal in the agonist pMHCII population than the null pMHCII population (grey = WT full length CD4 and blue = T1). In short, the right graph shows identification of those cells that are signaling in response to agonist pMHCII. This approach mitigated the need for subsequent gaBng in FlowJo to identify signaling cells (i.e., pCD3z positive) and allowed for background subtraction which could not be done in FlowJo. We used this approach for all analyses of pCD3z, pZAP-70, and pPLC γ 1 in this study.

The number of cells in these background-subtracted populations were divided by 10,000 (the number of events collected and analyzed) to calculate the percent of responding 5c.c7+ CD4+ 58a-b- T cell hybridomas, while the mean fluorescent intensity for the cells within these populations represent the signaling intensity.

Panel D. The graph on the left shows the mean fluorescence intensity (MFI) $\pm$ SEM for the positive signaling population from the right graph of panel C. We see in this example comparing a WT and T1 cell line, generated at the same time from the same parental 58a-b- T cell hybridoma population, that the T1 MFI is significantly greater than the WT. These intensity values represent one of the paired intensity values used in the main Fig 2B (Left graph), where we show the paired MFI analysis of responding populations from 5 independently generated sets of cell lines. Please note that these single MFI values are directly derived from the flow cytometry histograms average background subtraction. Figure 2B, and similar figures, therefore equate to a distribution of all of the histograms for the populations tested in a manner that we consider easier to digest than either overlaying all histograms or showing multiple panels individually. It also conserves more space. This is why we only showed representative flow cytometry histograms, rather than all histograms.

The graph on the right shows the % responders for the positive signaling population from the right graph of panel C. Specifically, the total number of cells that were determined to be signaling in response to agonist pMHCII was divided by 10,000 (the number of coupled cells collected by flow cytometry) to determine the percent responders. These values represent one of five sets of values used to determine the average normalized percent responders (all normalized to WT). There was no significant difference between these two populations in terms of percent responders.

Regarding graphing normalized values for the mean MFI for signaling intensity or the percent responders: in our first manuscript, we presented the individual MFI intensity values for matched pairs of cells as well as the actual percent responders per group. The feedback we received from colleagues on this presentation was that it was confusing, distracting, and otherwise hard to digest. It was suggested to us by multiple individuals that the normalized values would be preferable because it is easier and faster to understand. Upon reflection, we

agreed with this feedback because the normalized presentation with standard bars allows for the two key relevant questions to be quickly evaluated: 1. Are the mutants different than the control? 2. By how much? We have level the raw intensity values and well as the normalized intensity values in the version of record. Given the Reviewer's comments, we have now graphed the average % responders instead of normalized values in the figures, and level the normalized values in Table 3.

(4) The manuscript lacks Materials and Methods. It only refers to the previous paper, which is very unusual. Although most of the methods are the same, they should be mentioned here. Moreover, some of the mutants presented here were not generated in the previous study, as far as I understand. Perhaps the authors plan to include Materials and Methods during the revision...

Because we submitted this as a Research Advances article we followed the journal instructions to reference the Materials and Methods in our prior publication, upon which this work builds, as the methods used are the same. They are detailed in that study. We have now included a copy of the Materials and Methods for the eLife staff to determine how best to link with this manuscript. We have also included the gene sequences for the novel constructs used in this study. Thank you for pointing out the omission.

(5) Membrane rafts are a very controversial topic. I recommend the authors stick to the more consensual term "detergent resistant microdomains" in all cases/occurrences.

We agree this is a controversial topic with a variety of viewpoints. Because we are not experts in the field of membrane composition, we turned to the literature to inform our view of how best to refer to these membrane subdomains. In our reading, we found a 2006 meeting report from a Keystone symposium on lipid rafts and cell function authored by Linda Pike (PMID 16645198). At this meeting, a central focus was reaching a consensus on how best to refer to these domains. The consensus term agreed upon by this group was "membrane rafts". Specifically, we will quote from this report published in the Journal of Lipid Research, "Together, the discussions permitted the generation of a definition for "lipid rafts" in an ad hoc session on the final day of the meeting. All participants were invited to contribute to this effort, and the work product reflects the consensus of this broad-based group..... First and foremost, the term "lipid raft" was discarded in favor of the term "membrane raft." We chose to use the term "membrane raft" based on this consensus opinion.

(6) Last, but not least, the mechanistic explanation (beyond the independence of LCK binding) of the role of these motifs is very unclear at the moment.

We agree with this comment. One goal in making these results, and those in our prior study, available to the field at large is to provide evidence in support of our view that the dominant paradigm that is thought to explain the earliest events in T cell signaling needs re-evaluating. How T cell signaling is initiated in response to pMHCII is clearly more complex than is currently thought. However, our data is inconsistent with the dominant paradigm in which CD4 recruits Lck to TCR-CD3 to phosphorylate ITAMs to initiate signaling.

Reviewer #2 (Public Review):

Summary:

The paper by Kuhn and colleagues follows upon a 2022 eLife paper in which they identified residues in CD4 constrained by evolutionary purifying selection in placental mammals and then performed functional analyses of these conserved sequences. They showed that sequences distinct from the CXC "clamp" involved in recruitment of Lck have critical roles in TCR signaling, and these include a glycine-rich motif in the

transmembrane (TM) domain and the cysteine-containing juxtamembrane (JM) motif that undergoes palmitoylation, both of which promote TCR signaling, and a cytoplasmic domain helical motif, also involved in Lck binding, that constrains signaling. Mutations in the transmembrane and juxtamembrane sequences led to reduced proximal signaling and IL-2 production in a hybridoma's response to antigen presentation, despite retention of abundant CD4 association with Lck in the detergent-soluble membrane fraction, presumably mislocalized outside of lipid rafts and distal to the TCR. A major conclusion of that study was that CD4 sequences required for Lck association, including the CXC "clasp" motif, are not as consequential for CD4 co-receptor function in TCR signaling as the conserved TM and JM motifs. However, the experiments did not determine whether the functions of the TM and JM motifs are dependent on the Lck-binding properties of CD4 - the mutations in those motifs could result in free Lck redistributing to associate with CD4 in signaling-incompetent membrane domains or could function independently of CD4-Lck association. The current study addresses this specific question.

Using the same model system as in the earlier eLife paper (the entire methods section is a citation to the earlier paper), the authors show that truncation of the Lck-binding intracellular domain resulted in a moderate reduction in IL-2 response, as previously shown, but there was no apparent effect on proximal phosphorylation events (CD3z, Lck, ZAP70, PLCg1). They then evaluated a series of TM and JM motif mutations in the context of the truncated Lck-nonbinding molecule, and showed that these had substantially impaired co-receptor function in the IL-2 assay and reduced proximal signaling. The proximal signaling could be observed at high ligand density even with a MHC non-binding mutation in CD4, although there was still impaired IL-2 production. This result additionally illustrates that phosphorylation of the proximal signaling molecules is not sufficient to activate IL-2 expression in the context of antigen presentation.

Strengths:

The strength of the paper is the further clear demonstration that the classical model of CD4 coreceptor function (MHCII-binding CD4 bringing Lck to the TCR complex, for phosphorylation of the CD3 chain ITAMs and of the ZAP70 kinase) is not sufficient to explain TCR activation. The data, combined with the earlier eLife paper, further implicate the gly-rich TM sequence and the palmitoylation targets in the JM region as having critical roles in productive co-receptor-dependent TCR activation.

Weaknesses:

The major weakness of the paper is the lack of mechanistic insight into how the TM and JM motifs function. The new results are largely incremental in light of the earlier paper from this group as well as other literature, cited by the authors, that implicates "free" Lck, not associated with co-receptors, as having the major role in TCR activation. It is clear that the two motifs are important for CD4 function at low pMHCII ligand density. The proposal that they modulate interactions of TCR complex with cholesterol or other membrane lipids is an interesting one, and it would be worth further exploring by employing approaches that alter membrane lipid composition. The JM sequence presumably dictates localization within the membrane, by way of palmitoylation, which may be critical to regulate avidity of the TCR:CD4 complex for pMHCII or TCR complex allosteric effects that influence the activation threshold. Experiments that explore the basis of the mutant phenotype could substantially enhance the impact of this study.

We appreciate these thoughtful comments and suggestions. We will restate what we wrote in our preliminary response to the reviews to explain the scope of the current study:

To address comments about the limited scope of this study and referencing of the Methods section to our prior study, we would like to note that we submitted the current study via the

Research Advance mechanism. Our goal was to build upon the conclusions of our 2022 eLife publicaBon (PMID: 35861317) and address an unresolved quesBon from that study (as nicely summarized by Reviewer #2). In the current manuscript we present data from reducBonist experiments that were designed specifically for this purpose and, as noted by the reviewers, we provide answers to the quesBon being asked. We think that the Research Advance mechanism is an ideal opportunity to make these results available to the field given the stated purpose of such arBcles (for reference: “A Research Advance might use a new technique or a different experimental design to generate results that build upon the conclusions of the original research by, for example, providing new mechanis=c insights or extend the pathway under inves=ga=on...”). Now that we have provided evidence that CD4 does not recruit Lck to phosphorylate TCR-CD3 ITAMs in our system, nor do the GGXXG and (C/F)CV+C motifs play a role in enabling CD4 to regulate Lck proximity to TCR-CD3, we agree that it is important to form and test alternative hypotheses for how TCR-CD3 signaling is initiated.